

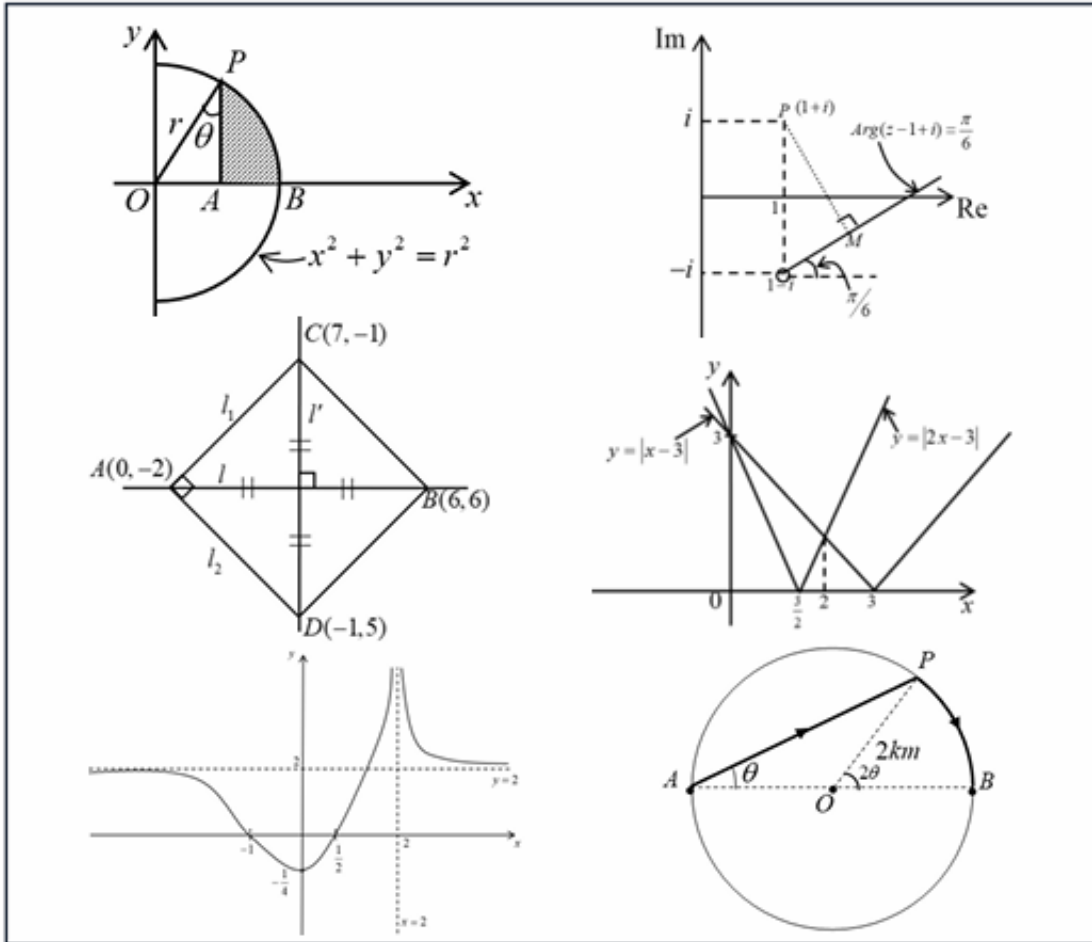


முல்லை பொறியியல் மற்றும் மருத்துவ பீட மாணவர் ஒன்றியம்
நடாத்தும்

DREAM WAY EXAMINATION 2026

இணைந்த கணிதம்-I,II

விடைகள்(SCHEME)



ORGANIZED BY:

ENGINEERING AND MEDICAL STUDENTS ASSOCIATION OF MULLAITIVU-EMSAM



01) $n=1$ க்கு

$1 \cdot 1 \cdot 3$

$2 \cdot 1 \cdot 5 (1+2) = 3 \times 2 \times 1 = 6$

$\therefore n=1$ க்கு முடிவு உண்மை.

$n=k$ க்கு உண்மை என்க.

$3^k < C(k+2)!$

$n=k+1$ க்கு

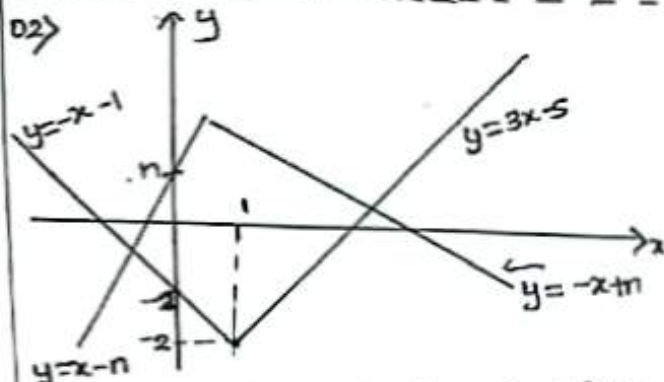
3^{k+1}

3^{k+3}

$< C(k+2)!(k+3)$

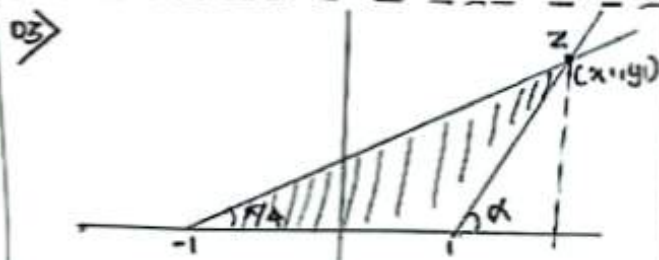
$< C(k+3)!$

$\therefore$ மணிதரத்திற்கு க்கு உண்மையான மைம்
மெய் $n \in \mathbb{Z}^+$ க்கும் முடிவு உண்மையானம்.



$|x - n| + 2|x - 1| = 3$ க்கு அகற்சு

மெய் $n=1$ க்கும்.



$\frac{y_1 - 0}{x_1 + 1} = 1$

$y_1 = x_1 + 1$

$2x_1 = 4$

$x_1 = 2$

$\frac{y_1 - 0}{x_1 - 1} = 3$

$y_1 = 3x_1 - 3$

$y_1 = 3$

$\therefore Z = 2 + 3$

மறுப்பாறு $= \frac{1}{2} \times 2 \times 3 = 3$ க்கும்.

02) $C(1+x(2x-1))^7$

${}^7C_0(x(2x-1))^0 + {}^7C_1(x(2x-1))^1 + \dots$

x க்கு க்கு 7C_0

x க்கு க்கு $-{}^7C_1 = -7$

$(x+1)(2x^2-x+1)$ க்கு x க்கு க்கு $=1$

03) $\lim_{x \rightarrow 0} \frac{\sin(\pi - \pi \cos x) x (\sqrt{x^2+1} + 1)}{(x^2+1) - 1}$

$\lim_{x \rightarrow 0} \frac{\sin(\pi - \pi \cos x) (\sqrt{x^2+1} + 1) (1 - \cos x) \cdot \pi}{x^2 (1 - \cos x) x \pi}$

$\lim_{x \rightarrow 0} \frac{\sin(\pi - \pi \cos x) (\sqrt{x^2+1} + 1) (1 - \cos^2 x) \pi}{(\pi - \pi \cos x) (1 + \cos x \cdot x^2)}$

$\frac{1 \times 2 \times 1^2 \times \pi}{(1+1)} = \pi$

04) $\int_{1/2}^{\sqrt{3}/2} x \left(\frac{1}{x} + \frac{1}{\sqrt{1-x^2}} \right) dx$

$\int_{1/2}^{\sqrt{3}/2} \left[1 + (1-x^2)^{-1/2} \right] x dx$

$\int_{1/2}^{\sqrt{3}/2} dx - \frac{1}{2} \int_{1/2}^{\sqrt{3}/2} (1-x^2)^{-1/2} (-2x) dx$

$\Rightarrow x - \frac{1}{2} \frac{(1-x^2)^{1/2}}{1/2} \Big|_{1/2}^{\sqrt{3}/2}$

$\Rightarrow \frac{\sqrt{3}}{2} - \frac{1}{2} - \left[\frac{1}{2} - \frac{\sqrt{3}}{2} \right]$

$\sqrt{3} - 1$ க்கும்.

05) $\frac{dy}{dx} = 3x^2 + \frac{9}{2\sqrt{x}}$

$\left(\frac{dy}{dx} \right)_{x=1} = 3 + \frac{9}{2} = 4$

$\frac{a}{2} = 1$

$a = 2$

$4x - y - 2 = 0, x = 1$

$4 - y - 2 = 0$

$y = 2$

$2 = 1 + a + b$

$a + b = 1$

$b = -1$

06) $\frac{y-1}{x-3} = m$

$y - 1 = m(x - 3)$

$mx - y + 1 - 3m = 0$

$1 = \frac{|1-3m|}{\sqrt{m^2+1}}$

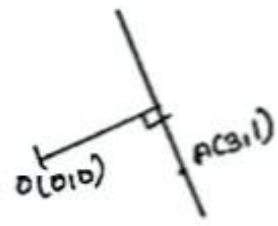
$m^2 + 1 = 1 - 6m + 9m^2$

$8m^2 - 6m = 0$

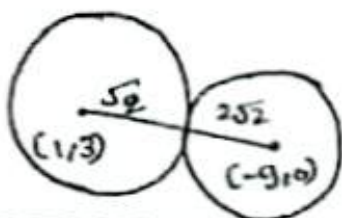
$2m(4m - 3) = 0$

$m = 0, m = \frac{3}{4}$

$y = 1, \frac{3}{4}x - y - \frac{5}{4} = 0$



09)



$$\sqrt{(1+9)^2 + 9} = 3\sqrt{2}$$

$$1 + 29 + 9 + 9 = 48$$

$$9^2 + 29 - 8 = 0$$

$$C_9 + 4(C_9 - 2) = 0$$

$$C_9 = -4, C_9 = 2 \quad C = 16 - 8 = 8$$

$$x^2 + y^2 + 8x + 8 = 0$$

$$x^2 + y^2 + 4x - 4 = 0$$

$$10) 2 \left[\frac{1}{2} (\cos 3x + \frac{\sqrt{3}}{2} \sin 3x) \right] = 2 \left[\frac{1}{2} \cos \theta + \frac{1}{2} \sin \theta \right]$$

$$3x - \frac{\pi}{3} = 2n\pi \pm (x - \frac{\pi}{4})$$

$$(+) \quad 2x = 2n\pi - \frac{\pi}{4} + \frac{\pi}{3}, x = n\pi + \frac{\pi}{12}$$

$$(-) \quad \therefore$$

$$4x = 2n\pi + \frac{\pi}{12}$$

$$x = \frac{1}{4} \left[2n\pi + \frac{\pi}{12} \right]$$

$$n \in \mathbb{Z}$$

11) a)

(i) $\alpha > 0, \Delta < 0$

$1 > 0, (2\lambda - 1)^2 - 4\lambda^2 < 0$

$4\lambda^2 - 4\lambda + 1 - 4\lambda^2 < 0$

$4\lambda > 1$

$\lambda > \frac{1}{4}$

———— (20)

$(1 - \lambda^2)[x^2 - (2\lambda - 1)x + \lambda^2] > 0$ if

$1 - \lambda^2 > 0$

and $\lambda > \frac{1}{4}$

$(1 - \lambda)(1 + \lambda) > 0$

$-1 < \lambda < 1$ and $\lambda > \frac{1}{4}$

$\frac{1}{4} < \lambda < 1$

———— (20)

ii) $x^2 - (2\lambda - 1)x + \lambda^2 = 0$

$\Delta = 4\lambda^2 - 4\lambda + 1 - 4\lambda^2 = 0$

$1 - 4\lambda$

$\lambda \leq \frac{1}{4}$

$\Delta \geq 0 \therefore$ அமையாமை — (15)

$\alpha + \beta = 2\lambda - 1$

$\alpha\beta = \lambda^2$

$(\alpha + \beta + 1)^2 = (\alpha + \beta)^2 + 2(\alpha + \beta) + 1$

$= (2\lambda - 1)^2 + 2(2\lambda - 1) + 1$

$= 4\lambda^2 - 4\lambda + 1 + 4\lambda - 2 + 1$

$= 4\lambda^2 = 4\alpha\beta$ — (15)

$(\alpha + \beta + 1)^2 = 4\lambda^2$

$\alpha + \beta + 1 = 2\lambda$

$\frac{\alpha + \beta + 1}{2} = \lambda$

$\frac{\alpha + \beta + 1}{2} \leq \frac{1}{4}$

$\alpha + \beta \leq -\frac{1}{2}$

———— (20)

(b) $x^4 - ax^3 + 3x^2 + b = \phi(x)(x-2)^2 + q(1-x)$

$x=2 \quad 16 - 8a + 12 + b = 0 - a$

$7a - b = 28$

கூடுதல் வகையிலு

$4x^3 - 3ax^2 + 6x = \phi(x)2(x-2) + (x-2)^2\phi'(x) + q$

$x=2$

$32 - 12a + 12 = 0 + 0 - a$

$44 = 11a$

$a=4$

———— (30)

$f(x) = x^4 - 4x^3 + 3x^2$

$f(x) + f'(x) + 4x$

$x^4 - 9x^2 + 10x$

$(x-2)(x^2 + 2x - 5)$

$(x-2)(x+1-\sqrt{6})(x+1+\sqrt{6})$

———— (30)

12) (a) A B

உயர் 3 4

தாழ் 3 4

(i) ${}^{17}C_5 = 6188$ — (15)

(ii) ${}^{17}C_5 - {}^{10}C_5 - {}^7C_5$

$6188 - 252 - 21 = 5915$ — (20)

(iii) ${}^{17}C_8 - {}^6C_5 - {}^{11}C_5$

$6188 - 6 - 462 = 5720$ — (15)

iv) $\text{பொதுக்கூறு} = (\text{இருபடி இயல்பு} + \text{இருபடி இயல்பு} - \text{இருபடி இயல்பு}) - \text{இருபடி இயல்பு}$

$6188 - (273 + 468) - 21$

$= 5468$ — (20)

b) $\frac{1}{(2r+1)^2} - \frac{1}{(2r+3)^2}$

$\frac{4r^2 + 12r + 9 - (4r^2 - 4r + 1)}{(2r+1)^2(2r+3)^2}$

$\frac{16r+8}{(2r+1)^2(2r+3)^2}$

$\frac{8(2r+1)}{(2r+1)^2(2r+3)^2} = 8ur$ — (20)

$8ur = f(r) - f(r+2)$

$r=1 \quad 8u_1 = f(1) - f(3)$

$r=2 \quad 8u_2 = f(2) - f(4)$

$r=3 \quad 8u_3 = f(3) - f(5)$

$r=n-2 \quad 8u_{n-2} = f(n-2) - f(n)$

$r=n-1 \quad 8u_{n-1} = f(n-1) - f(n+1)$

$r=n \quad 8u_n = f(n) - f(n+2)$

$\sum_{r=1}^n 8ur = f(1) + f(2) - f(n+1) - f(n+2)$

$\sum_{r=1}^n ur = \frac{1}{1} + \frac{1}{9} - \frac{1}{(2n+1)^2} - \frac{1}{(2n+3)^2}$ — (30)

$= \frac{5}{36} - \frac{1}{8(2n+1)^2} - \frac{1}{8(2n+3)^2}$

$n \rightarrow \infty \quad \lim_{n \rightarrow \infty} \frac{5}{36} - \frac{1}{8(2n+1)^2} - \frac{1}{8(2n+3)^2}$

$\alpha = \frac{5}{36} \therefore$ அதன் மதிப்பாகும்

$\sum_{r=1}^{\infty} ur = \frac{5}{36}$ — (10)

$$V_n = \sum_{r=1}^n u_r - \sum_{r=1}^{n-1} u_r$$

$$= \frac{5}{36} - \left[\frac{5}{36} - \frac{1}{8(2n-1)^2} - \frac{1}{8(2n+1)^2} \right]$$

$$= \frac{1}{8(2n-1)^2} + \frac{1}{8(2n+1)^2}$$

$$\lim_{n \rightarrow \infty} \frac{n^n}{8(2n-1)} + \frac{n^2}{8(2n+1)^2}$$

$$\frac{1}{8(2-\frac{1}{n})^2} + \frac{1}{8(2+\frac{1}{n})^2}$$

$$= \frac{1}{32} + \frac{1}{32}$$

$$= \frac{1}{16} \quad \text{--- (20)}$$

13. (a)

$$\begin{pmatrix} a & 0 & a-1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ a & a-1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a+0+a-1 & a+0+0 \\ 0+a+0 & 0+a-1+0 \end{pmatrix}$$

$$\begin{pmatrix} 2a-1 & a \\ a & a-1 \end{pmatrix} = \begin{pmatrix} b & b-2 \\ b-2 & b-3 \end{pmatrix}$$

$$2a-1=b \quad a-1=b-3$$

$$2a-b=1 \quad \text{--- (1)} \quad b-a=2 \quad \text{--- (2)}$$

$$\textcircled{1} \text{ } \textcircled{2} \Rightarrow a=3, b=5 \quad \text{--- (30)}$$

$$C = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad C^{-1} = \frac{1}{10-9} \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}$$

$$C^{-1} = \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}$$

$$C+C^{-1} \quad \text{--- (20)}$$

$$\begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix} + \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix} = 7 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 7I$$

open $\rightarrow$ (10)

(b) $z = x+iy$ or

$$z+i = x+(y+1)i$$

$$|z+i| = \sqrt{x^2 + (y+1)^2}$$

$$|z+i|^2 = x^2 + y^2 + 2y + 1$$

$$= |z|^2 + 2\operatorname{Im}z + 1 \quad \text{(OR)}$$

$$z+i|^2 = |z+i| \overline{|z+i|}$$

$$= (z+i)(\overline{z+i})$$

$$= z\bar{z} - i(z-\bar{z}) + 1$$

$$= |z|^2 + 2\operatorname{Im}z + 1$$

$$\frac{1}{(z+i)(z-i)} \quad [\bar{z} = x-iy]$$

$$\frac{z-i(y+1)}{|z+i|^2}$$

$$\therefore \operatorname{Im}\left(\frac{1}{z+i}\right) = \frac{-(1+\operatorname{Im}z)}{|z+i|^2} \quad \text{--- (10)}$$

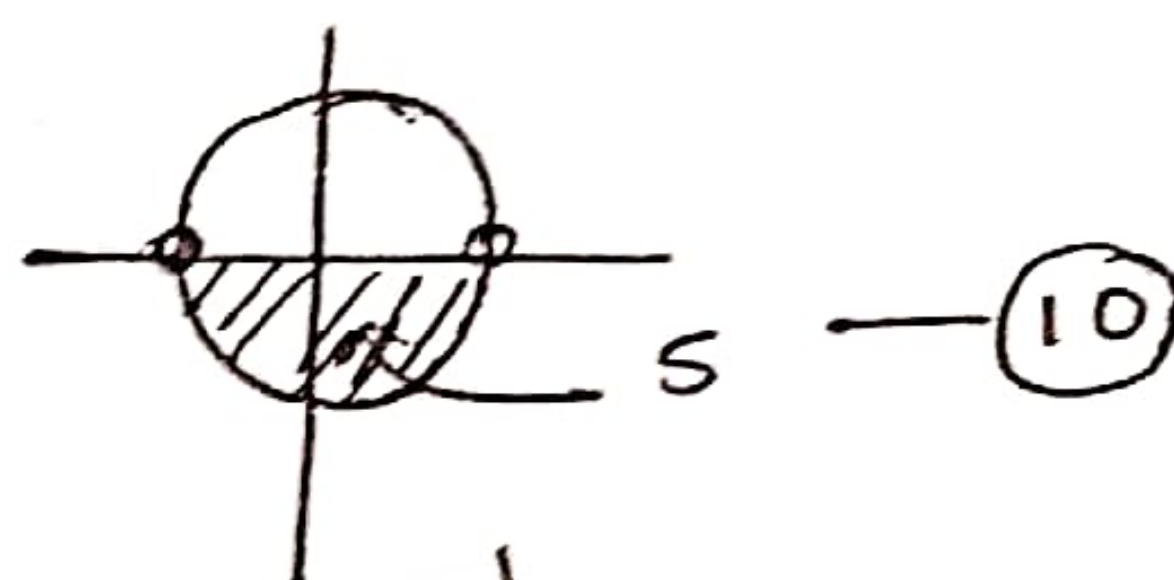
$$\operatorname{Im}\left(\frac{1}{z+i}\right) = -\frac{1}{2}$$

$$\Rightarrow \frac{-1(1+\operatorname{Im}z)}{|z+i|^2} = -\frac{1}{2}, \quad z \neq i$$

$$\Rightarrow 2+2\operatorname{Im}z = |z|^2 + 2\operatorname{Im}z + 1, \quad z \neq i$$

$$= |z|^2 = 1, \quad |z| \neq 1$$

$$\Rightarrow |z|=1, \quad z \neq i \quad \text{--- (16)}$$



$$\cos\theta + \sin\theta = \frac{1}{\sqrt{2}}$$

$$\sqrt{2} \left[\frac{1}{\sqrt{2}} \cos\theta + \frac{1}{\sqrt{2}} \sin\theta \right] = \frac{1}{\sqrt{2}}$$

$$\cos(\theta - \pi/4) = 1/2$$

$$\theta - \pi/4 = 2n\pi \pm \frac{2\pi}{3}$$

$$n=0 \quad \theta - \pi/4 = \pm 2\pi/3$$

$$\theta_1 = -2\pi/3 + \pi/4, \quad 2\pi/3 + \pi/4 \text{ are not possible}$$

$$\theta_2 = -\frac{2\pi}{3} + \frac{\pi}{4}$$

$$= -\frac{5\pi}{12}$$

$$\therefore z = \cos \frac{5\pi}{12} + i \sin \left(-\frac{5\pi}{12} \right)$$

$$z = \cos \frac{5\pi}{12} - i \sin \frac{5\pi}{12} \quad \text{--- (15)}$$

c) Given $z^n = \cos n\theta + i \sin n\theta$

$$z^n = \cos n\theta + i \sin n\theta$$

$$\frac{1}{z^n} = \bar{z}^n = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta - i \sin n\theta$$

$$z^n + \frac{1}{z^n} = 2 \cos n\theta$$

$$z^n - \frac{1}{z^n} = 2i \sin n\theta$$

$$z + \frac{1}{z} = 2 \cos \theta$$

$$\left(z + \frac{1}{z} \right) = 2 \cos \theta$$

$$\left(z + \frac{1}{z}\right)^6 = 2^6 \cos^6 \theta = 64 \cos^6 \theta$$

$$\left(z - \frac{1}{z}\right)^6 = 2^6 \sin^6 \theta = -64 \sin^6 \theta$$

$$\left(z + \frac{1}{z}\right)^6 = z^6 + 6z^4 + 15z^2 + 20 + \frac{15}{z^2} + \frac{6}{z^4} + \frac{1}{z^6}$$

$$= \left(z^6 + \frac{1}{z^6}\right) + 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) + 20$$

$$= 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20 \quad \text{--- (1)}$$

$$\left(z - \frac{1}{z}\right)^6 = z^6 - 6z^4 + 15z^2 - 20 + \frac{15}{z^2} - \frac{6}{z^4} + \frac{1}{z^6}$$

$$= z^6 + \frac{1}{z^6} - 6\left(z^4 + \frac{1}{z^4}\right) + 16\left(z^2 + \frac{1}{z^2}\right) - 20$$

--- (25)

(14)(a) $f(x) = \frac{(2x-1)^2}{(x-2)(x^2+1)}$

$$f'(x) = \frac{(x-2)(x^2+1)2(2x-1) - (2x-1)^2(3x^2-4x+1)}{(x-2)^2(x^2+1)^2}$$

$$= \frac{-(2x-1)(x+1)(2x^2-5x+7)}{(x-2)^2(x^2+1)^2} \quad \text{--- (20)}$$

$$y = \frac{(2x-1)^2}{(x-2)(x^2+1)}$$

$$x=0 \Rightarrow y = -1/2, y=0 \Rightarrow x=1/2$$

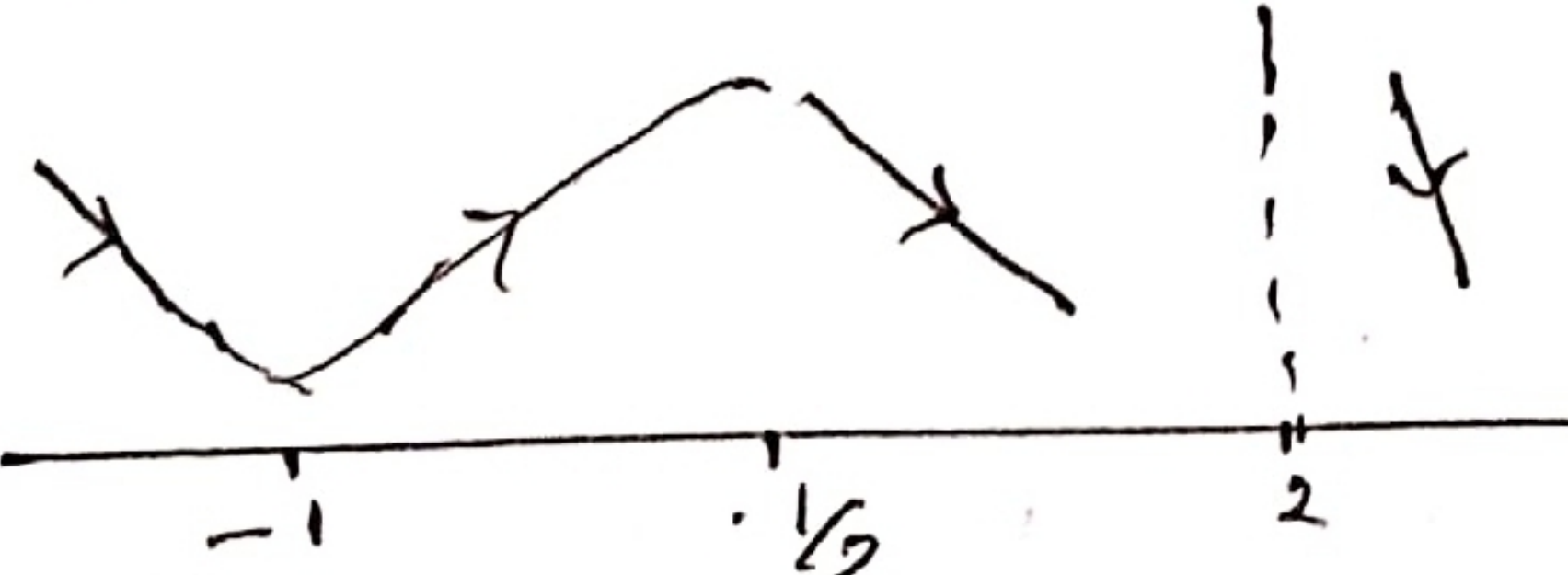
$x=2$ is a vertical asymptote

$$\frac{dy}{dx} = 0$$

$$\frac{(2x-1)(x+1)(2x^2-5x+7)}{(x-2)^2(x^2+1)^2} = 0$$

$$x = 1/2, x = -1$$

$f(x)$



$$x = -1 \Rightarrow y = -3/2$$

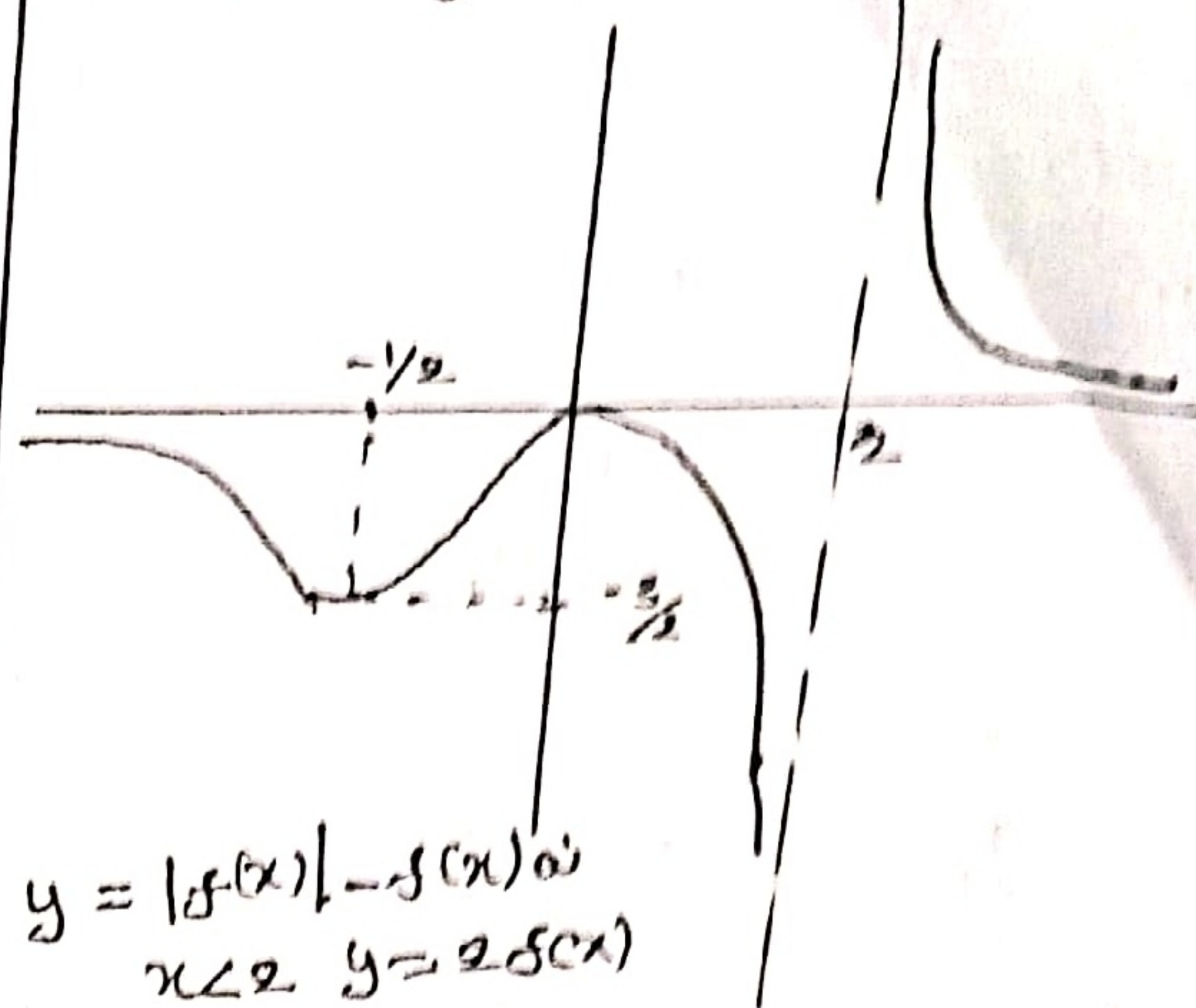
$$x = 1/2 \Rightarrow y = 0$$

$(-1, -3/2)$ is a local minimum

$(1/2, 0)$ is a local maximum

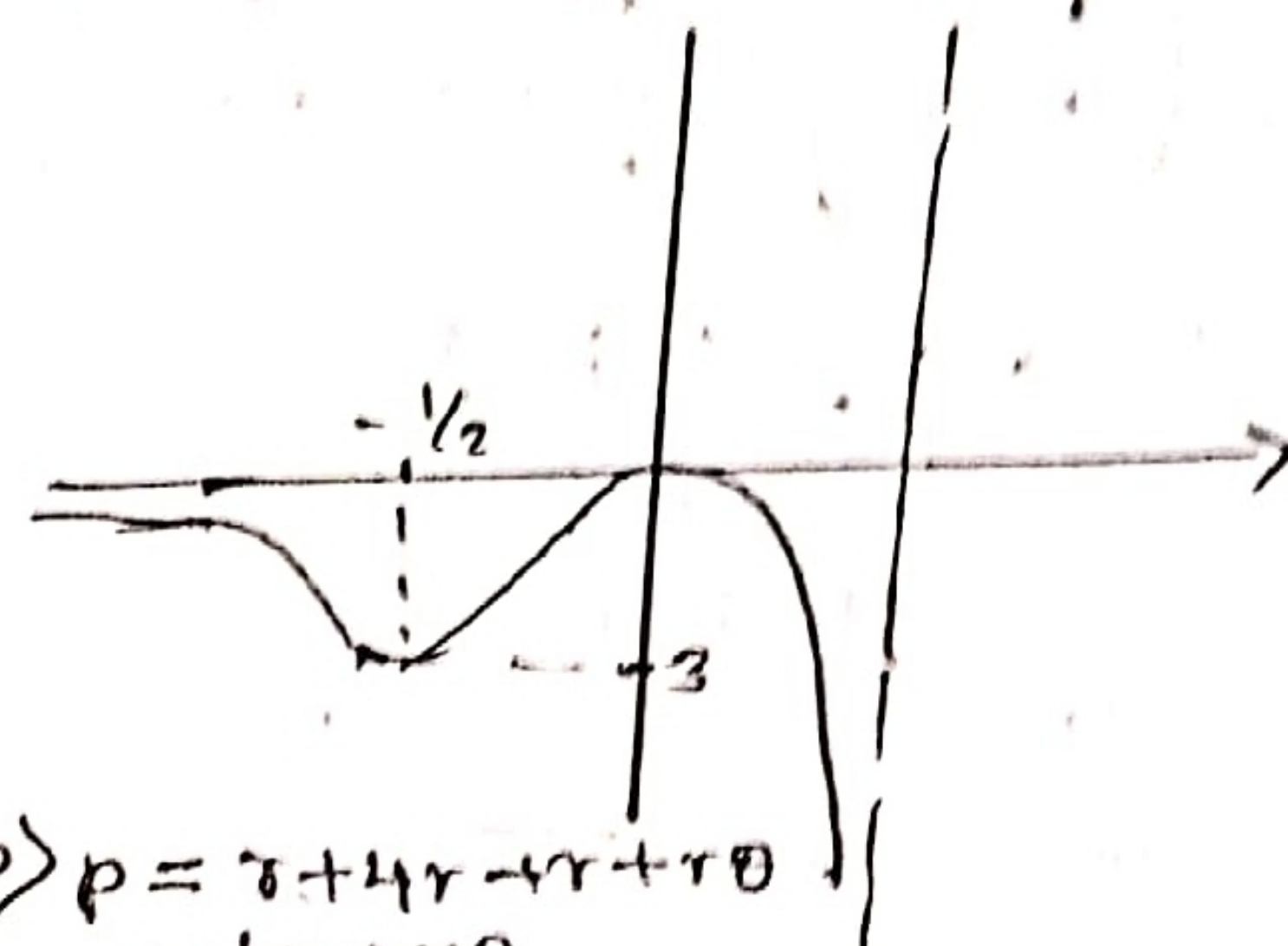
$$y = \frac{x^2(2 - \frac{1}{x})^2}{x^3(1 - \frac{2}{x})(1 + \frac{1}{x^2})}$$

$$x \rightarrow \pm \infty \Rightarrow y \rightarrow 0$$



$$y = |f(x)| - f(x) \text{ or } x < 2 \Rightarrow y = 2f(x)$$

$$x > 2 \Rightarrow y = 0$$



b) $p = r + 4r - r + r\theta$

$$= 6r + r\theta$$

$$6r + r\theta = 15$$

$$r\theta = 15 - 6r$$

$$\theta = \frac{15}{r} - 6$$

$$A = 2r \times r + \frac{1}{2} \theta r^2$$

$$= 2r^2 + \frac{1}{2} r^2 \left(\frac{15}{r} - 6\right)$$

$$= \frac{15r}{2} - r^2$$

$$\frac{dA}{dr} = \frac{15}{2} - 2r$$

$$\frac{dA}{dr} = 0$$

$$r = \frac{15}{4}$$

$$\frac{15}{4} \quad \therefore r = \frac{15}{4} \text{ is a maximum}$$

$$A = \frac{15}{2} \times \frac{15}{4} - \left(\frac{15}{4}\right)^2$$

$$= \frac{225}{8} - \frac{225}{16}$$

$$= \frac{225}{16} \text{ cm}^2$$

--- (60)

(15)(a)

$$\frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)}$$

$$1 = A(x+1) + B(x-1)$$

$$[x] 0 = A+B$$

$$[] 1 = A-B$$

$$A = \frac{1}{2}, B = \frac{1}{2} \quad \text{--- (15)}$$

$$f(x) = \frac{1}{(x^2+1)(x^2-1)} = \frac{1}{2(x^2-1)} - \frac{1}{2(x^2+1)}$$

$$= \frac{1}{4(x-1)} - \frac{1}{4(x+1)} - \frac{1}{2(x^2+1)}$$

$$\int f(x) dx = \int \frac{1}{4(x-1)} dx - \int \frac{1}{4(x+1)} dx - \int \frac{1}{2(x^2+1)} dx$$

$$= \frac{1}{4} \ln|x-1| - \frac{1}{4} \ln|x+1| - \frac{1}{2} \tan^{-1} x + C \quad \text{--- (35)}$$

$$x = \sin \theta \Rightarrow dx = \cos \theta d\theta$$

θ	θ	$\pi/6$
x	0	$1/2$

$$\frac{1}{2} \int_0^{\pi/6} \frac{1}{(x^2+1)\cos \theta} \cdot \frac{dx}{\cos \theta} = \frac{1}{2} \int_0^{\pi/6} \frac{dx}{(x^2+1)(1-x^2)}$$

$$= -\frac{1}{2} \int_0^{\pi/6} \frac{dx}{(1+x^2)(x^2-1)}$$

$$= -\frac{1}{2} \int_0^{\pi/6} f(x) dx$$

$$= \left[\frac{1}{4} \ln|x-1| - \frac{1}{4} \ln|x+1| - \frac{1}{2} \tan^{-1}(x) \right]_0^{\pi/6}$$

$$= \left[\frac{1}{4} (\ln \frac{1}{2}) - \ln(1) \right] - \frac{1}{4} \left[\ln \frac{3}{2} - \ln(1) \right] - \frac{1}{2} \left[\tan^{-1}(\frac{1}{2}) - \tan^{-1}(0) \right]$$

$$= \left[\frac{1}{4} \ln(\frac{1}{3}) - \frac{1}{2} \tan^{-1}(\frac{1}{2}) \right]$$

$$\frac{1}{4} \ln 3 + \frac{1}{2} \tan^{-1}(\frac{1}{2})$$

$$(b) I = \frac{\pi/6}{2} \int_0^{\pi/6} (\pi/6 - x) \sin(3(\pi/6 - x)) \cos^2(\pi/6 - x) dx$$

$$= \frac{\pi/6}{2} \int_0^{\pi/6} (\pi/6 - x) \sin(3(\pi/6 - x)) \cos^2(\pi/6 - x) dx$$

$$\frac{\pi/6}{2} \int_0^{\pi/6} (\pi/6 - x) \cos 3x \cdot \sin 3x \cdot \cos^2 3x dx$$

$$2I = \frac{\pi/6}{2} \int_0^{\pi/6} \frac{\pi/6 \cos 3x \sin 3x}{\cos^4 3x + \sin^4 3x} dx$$

$$I = \frac{\pi}{12} \int_0^{\pi/6} \frac{\sin 3x \cos 3x}{\cos^4 3x + \sin^4 3x} dx$$

$$\sin^4 3x + \cos^4 3x = (\sin^2 3x + \cos^2 3x)^2 - 2\sin^2 3x \cos^2 3x$$

$$= 1 - \frac{1}{2} \sin^2 6x$$

$$= \frac{1}{2} + \frac{1}{2} \cos^2 6x$$

$$I = \frac{\pi}{12} \int_0^{\pi/6} \frac{\frac{1}{2} \sin 6x}{\frac{1}{2} + \cos^2 6x} dx$$

$$\text{Let } t = \cos 6x$$

$$I = \frac{\pi}{12} \int_1^{-1} \frac{1}{1+t^2} \left(-\frac{dt}{t} \right)$$

$$= \frac{\pi}{12} \tan^{-1} x \Big|_{-1}^1$$

$$= \frac{\pi}{12} \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right)$$

$$= \frac{\pi}{12} \cdot \frac{\pi}{2}$$

$$= \frac{\pi^2}{144} = \left(\frac{\pi}{12} \right)^2 \quad \text{--- (40)}$$

$$(c) \int x^3 \sin 4x dx$$

$$= -\frac{\cos 4x}{4} x^3 - \int -\frac{\cos 4x}{4} 3x^2 dx$$

$$= \frac{3}{4} \left[\frac{\sin 4x}{4} x^2 - \int \frac{\sin 4x}{2} x dx \right]$$

$$+ \frac{3}{4} \left[\frac{\sin 4x}{4} x^2 - \frac{1}{2} \left[-\frac{\cos 4x}{4} x - \int -\frac{\cos 4x}{4} dx \right] \right]$$

$$+ \frac{\sin 4x}{16} \Big] + C$$

$$\text{--- (30)}$$

$$2b. \quad x = \frac{2 + 3\sin\theta}{2}, \quad \sin\theta = \frac{2x-2}{3}$$

$$y = \frac{5 + (3\cos\theta + 1)2\cos\theta}{2} = \frac{2y-6}{3}$$

$$\sin^2\theta + \cos^2\theta = 1$$

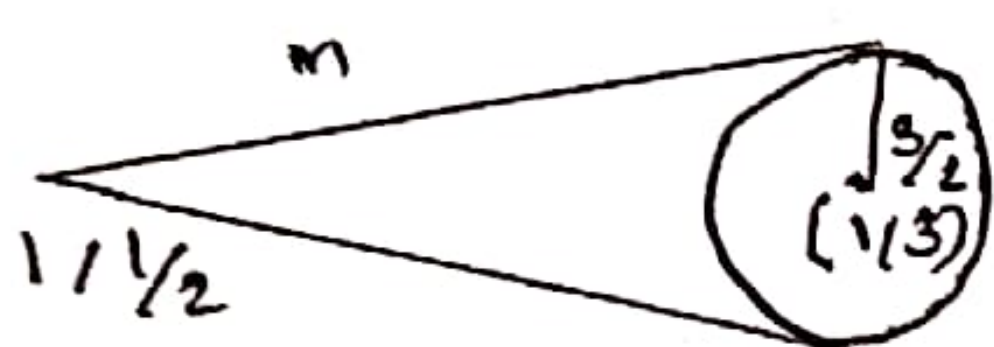
$$\left(\frac{2x-2}{3}\right)^2 + \left(\frac{2y-6}{3}\right)^2 = 1$$

$$4x^2 + 4y^2 - 8x - 24y + 31 = 0$$

$$x^2 + y^2 - 2x - 6y + \frac{31}{4} = 0$$

மையம் $(1, 3)$

$$r = \sqrt{1 + 9 - \frac{31}{4}} = \frac{3}{2} \quad (20)$$



$$\frac{y - 1/2}{x - 1} = m$$

$$y - 1/2 = mx - m$$

$$mx - y + 1/2 - m = 0$$

$$\frac{|m - 3 + 1/2 - m|}{\sqrt{m^2 + 1}} = \frac{3}{2}$$

$$9m^2 = 16$$

$$m = \pm \frac{4}{3}$$

$$\frac{5}{\sqrt{m^2 + 1}} = \frac{3}{2}$$

$$m = \frac{4}{3}$$

$$\frac{4}{3}x - y + \frac{1}{2} - \frac{4}{3} = 0$$

$$4x - 3y + \frac{3}{2} - 4 = 0$$

$$8x - 6y + 3 - 8 = 0$$

$$8x - 6y - 5 = 0$$

$$m = -\frac{4}{3}$$

$$-\frac{4}{3}x - y + \frac{1}{2} + \frac{4}{3} = 0$$

$$-4x - 3y + \frac{3}{2} + 4 = 0$$

$$-8x - 6y + 3 + 8 = 0$$

$$8x + 6y - 11 = 0 \quad (50)$$

DE வரையறுக்கப்பட்ட நேரிடையின் சமன்பாடு.

$$2x^2 + y^2 - (x+2) - 3(y+4) + \frac{31}{4} = 0$$

$$x \cdot 1 + y \cdot \frac{1}{2} - (1+2) - 3(\frac{1}{2}+4) + \frac{31}{4} = 0$$

$$-10y + 21 = 0 \quad (20)$$

$S=0$, DE வரையறுக்கப்பட்ட நேரிடையின் சமன்பாடு.

2-வது வரையறுக்கப்பட்ட நேரிடையின் சமன்பாடு

$$x^2 + y^2 - 2x - 6y + \frac{31}{4} + \lambda(21 - 10y) = 0$$

கிடைசு $(1, 3)$ மையம் மையம்.

$$1 + 9 - 2 - 18 + \frac{31}{4} + \lambda(21 - 30) = 0$$

$$10 - 20 + \frac{31}{4} - 9\lambda = 0$$

$$\lambda = -\frac{1}{4}$$

$$x^2 + y^2 - 2x - 6y + \frac{31}{4} - \frac{1}{4}(21 - 10y) = 0$$

$$x^2 + y^2 - 2x - 6y + \frac{31}{4} - \frac{21}{4} + \frac{5y}{2} = 0$$

$$2x^2 + 2y^2 - 4x - 7y + 5 = 0 \quad (30)$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$2(-1)(g) + 2(-7/4) = c + 5/2 = 0$$

$$-4g - 7f = 2c + 5 \quad (1)$$

$$2(-1)g + 2(-3)f = c + 31/4$$

$$-8g - 24f = c + 31 \quad (2)$$

$$(-1/2, 0)$$

$$\frac{1}{4} + 0 - g + 0 + c = 0$$

$$c = g - \frac{1}{4}$$

$$\Rightarrow g = \frac{17}{10}, \quad f = -\frac{21}{10}, \quad c = \frac{29}{20}$$

$$\therefore x^2 + y^2 + \frac{17}{5}x - \frac{21}{5}y + \frac{29}{20} = 0 \quad (30)$$

$$(17) f(x) = 16(\cos^4 x - \cos^2 x) + 1$$

$$16[\cos^2 x (\cos^2 x - 1)] + 1$$

$$-16\cos^2 x \sin^2 x + 1$$

$$-4\sin^2 2x + 1$$

$$-2(1 - \cos 4x) + 1$$

$$f(x) = 2\cos 4x - 1$$

$$A=2, \quad B=4, \quad C=-1, \quad \alpha=0 \quad (20)$$

$$f(x/2) = 2\cos(2x-1)$$

$$-1 \leq \cos 2x \leq 1$$

$$-2 \leq 2\cos 2x \leq 2$$

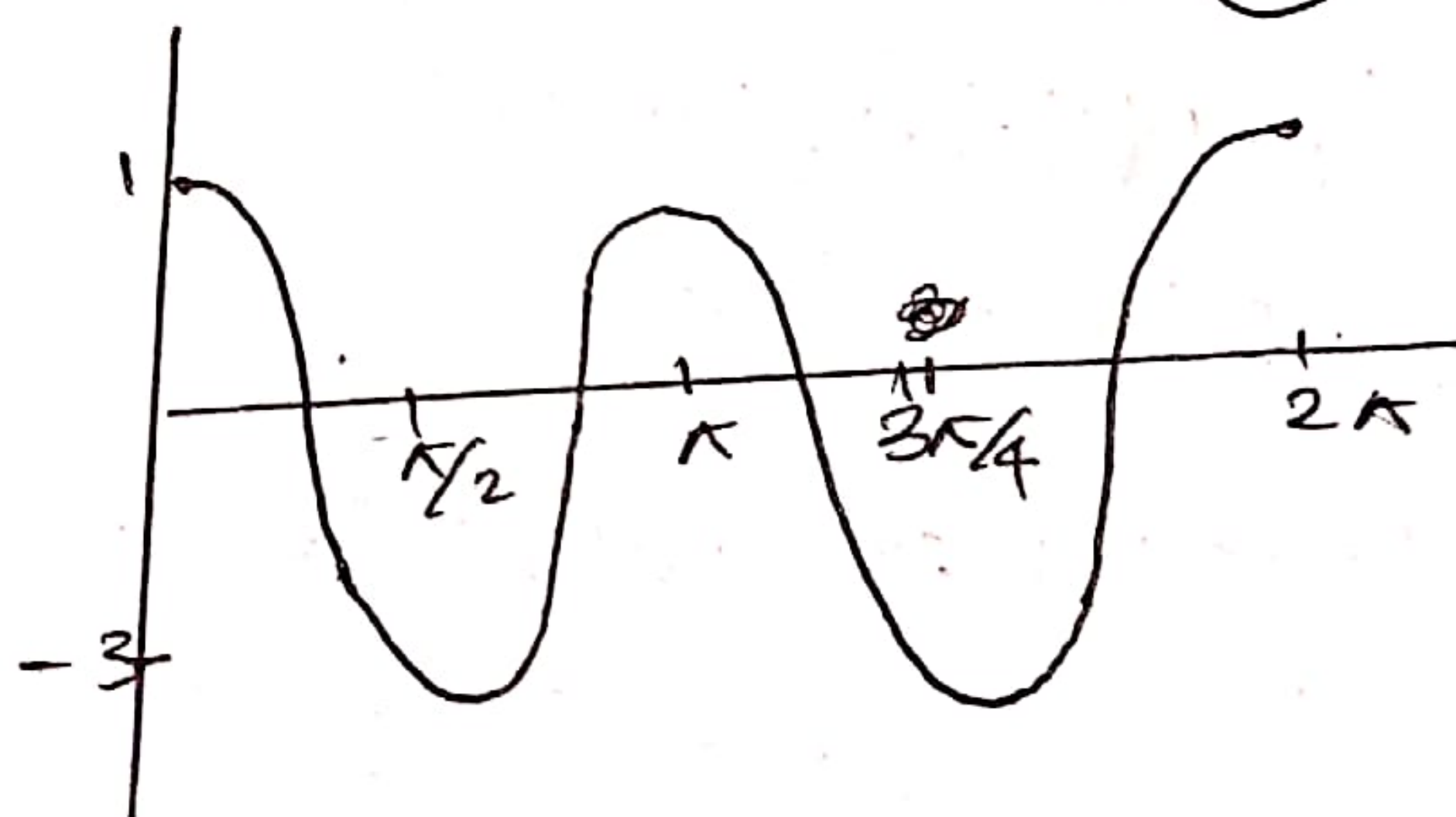
$$-3 \leq 2\cos 2x \leq 2$$

$$-3 \leq 2\cos 2x - 1 \leq 1$$

$$-3 \leq f(x/2) \leq 1$$

$$y = 2\cos 2x - 1$$

(30)



$$16 \cos^4 \frac{x}{2} + 16(1 - \cos^2 \frac{x}{2}) = k + 15$$

$$16 \cos^4 \frac{x}{2} + 16 - 16 \cos^2 \frac{x}{2} = k + 15$$

$$16 \cos^4 \frac{x}{2} - 16 \cos^2 \frac{x}{2} + 1 = k$$

$$y = k \quad \text{--- (20)}$$

$$(i) k = -3$$

$$(ii) k = 1$$

$$(iii) -3 < k < 1$$

$$\text{--- (10)}$$

b) ΔABC

$$\frac{AO}{\sin \alpha} = \frac{BO}{\sin(\alpha + \theta)}$$

$$AO = \frac{2 \cdot BO \sin \alpha}{3 \sin(\alpha + \theta)}$$

ΔCOB

$$\frac{CO}{\sin \beta} = \frac{BO}{\sin(\theta + \beta)}$$

$$CO = \frac{BO \sin \beta}{3 \sin(\theta + \beta)}$$

$$\frac{2 \cdot BO \sin \alpha}{3 \sin(\alpha + \theta)} = \frac{BO \sin \beta}{3 \sin(\theta + \beta)}$$

$$\frac{2 \sin \alpha \sin(\theta + \beta)}{3 \sin(\alpha + \theta)} = \frac{\sin \beta \sin(\alpha + \theta)}{3 \sin(\theta + \beta)}$$

$$2 \sin \alpha \sin(\theta + \beta) = \sin \beta \sin(\alpha + \theta)$$

$$\cot \theta = \cot \alpha - 2 \cot \beta \quad \text{--- (30)}$$

$$\cos^2 \frac{x}{2} + \sin^2 \frac{y}{2} = \frac{\pi}{2} \quad \text{--- (1)}$$

$$\tan^2 \frac{x}{2} - \tan^2 \frac{y}{2} = \frac{\pi}{6} \quad \text{--- (2)}$$

$$\frac{\pi}{2} - \sin^2 \frac{x}{2} + \sin^2 \frac{y}{2} = \frac{\pi}{2}$$

$$\sin^2 \frac{x}{2} = \sin^2 \frac{y}{2}$$

$$\frac{x}{2} = \frac{y}{2}$$

$$x = y$$

$$\tan^2 \frac{x}{2} - \tan^2 \frac{y}{2} = \frac{\pi}{6}$$

$$x = y$$

$$\therefore \theta = \frac{\pi}{6}$$

$$\therefore \text{Angle } \theta \text{ is } \frac{\pi}{6} \quad \text{--- (21)}$$



$$mu + \lambda mu_1 = \frac{mu}{1+\lambda} + \lambda m v_1$$

$$u + \lambda u_1 = \frac{u}{1+\lambda} + \lambda v_1$$

$$\frac{\lambda}{1+\lambda} u = \lambda(v - u_1)$$

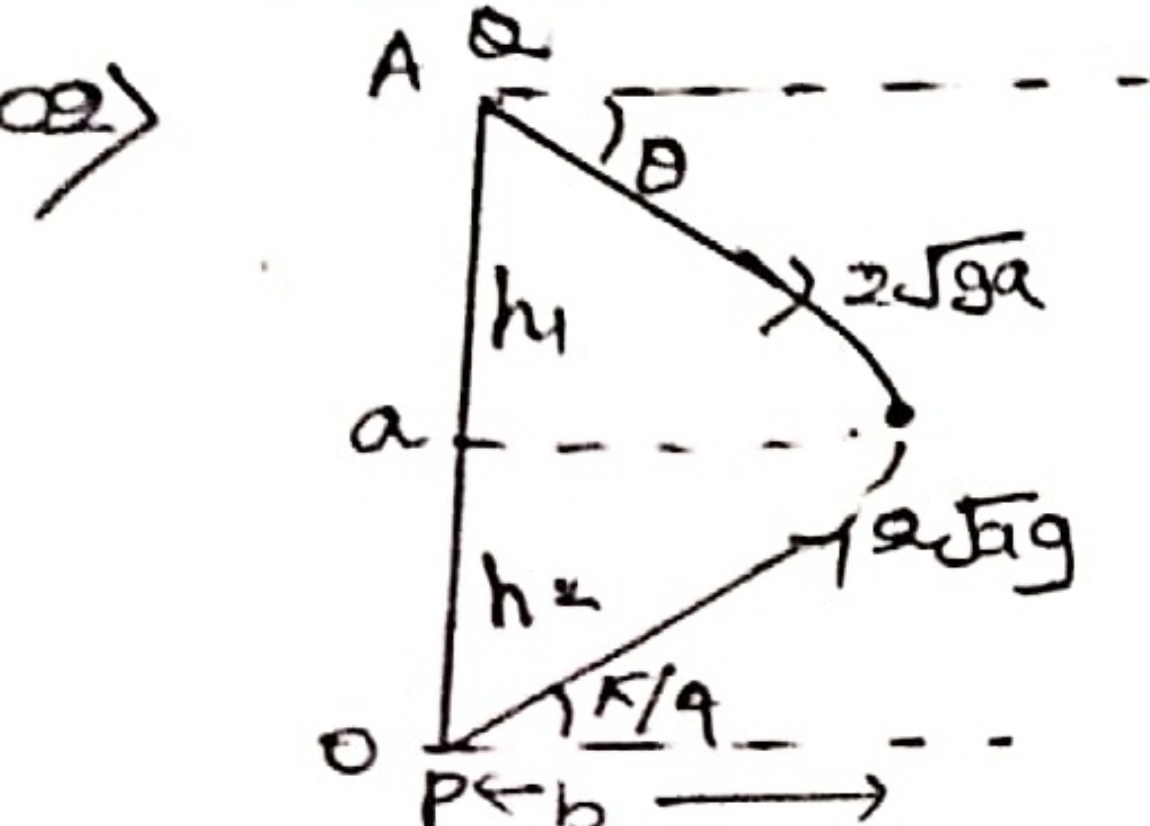
$$v_1 - u_1 = \lambda(v - u_1)$$

$$v_1 - u_1 = \frac{u}{1+\lambda} \quad \text{--- (1)}$$

$$v_1 - \frac{u}{1+\lambda} = \lambda(u - u_1)$$

$$v_1 + u_1 = \frac{(\lambda+2)u}{(1+\lambda)} \quad \text{--- (2)}$$

$$\text{(1) + (2)} \Rightarrow v_1 = \frac{(\lambda+3)u}{2(1+\lambda)}$$



$$A \rightarrow B \quad h_1 = 2\sqrt{ga} \sin \theta t + \frac{1}{2}gt^2 \quad \text{--- (1)}$$

$$O \rightarrow B \quad h_2 = \sqrt{ga} \sin \pi/4 - \frac{1}{2}gt^2 \quad \text{--- (2)}$$

$$a = (2\sqrt{ga} \sin \theta + \sqrt{ga})t \quad \text{--- (3)}$$

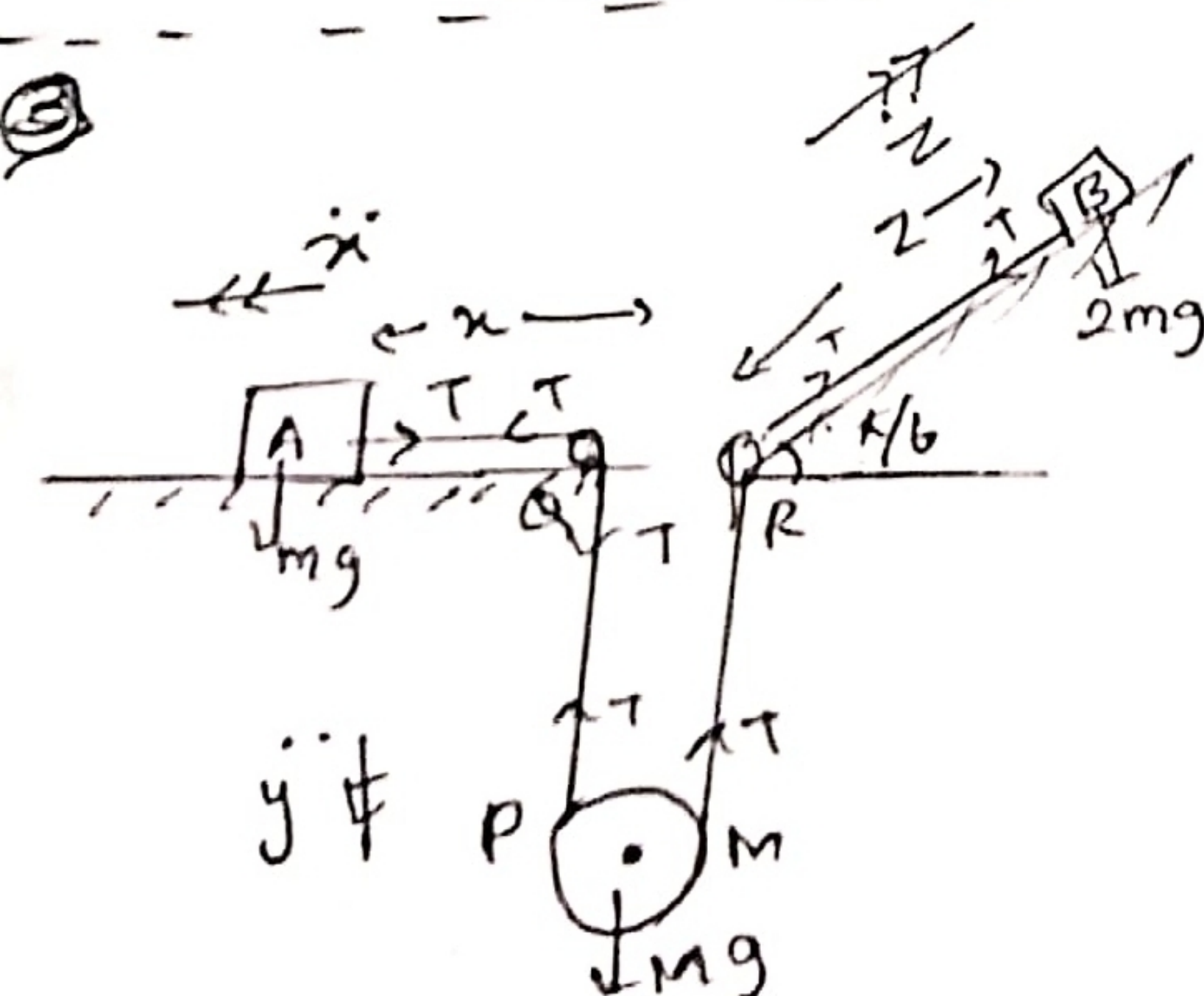
$$AB \rightarrow b = 2\sqrt{ga} \cos \theta t \quad \text{--- (4)}$$

$$b = \sqrt{2ga} \cos \pi/4 t \quad \text{--- (5)}$$

$$\cos \theta = \frac{1}{\sqrt{2}}, \quad \theta = \pi/4$$

$$\text{(3)} \Rightarrow a = \left(\frac{2\sqrt{ga}\sqrt{3}}{2} + \sqrt{ga} \right)t$$

$$t = \frac{1}{(\sqrt{3}+1)}\sqrt{\frac{g}{a}}$$



$$x + 2y + z = l$$

$$\dot{x} + 2\dot{y} + \dot{z} = 0$$

$$\ddot{x} + 2\ddot{y} + \ddot{z} = 0$$

$$\lambda \ddot{x} - T = m\ddot{x}$$

$$-T/m = \ddot{x} \quad \text{--- (1)}$$

$$P \rightarrow Mg - 2T = M\ddot{y}$$

$$g - \frac{2T}{M} = \ddot{y}$$

$$2g - \frac{4T}{M} = 2\ddot{y}$$

$$T_B = -T - 2mg \sin \pi/6 = 2m\ddot{z}$$

$$-\frac{T}{2m} - g/2 = \ddot{z} \quad \text{--- (3)}$$

$$P = F \times V$$

$$40 \times 10^3 = F \times 10$$

$$F = 4000 \text{ N}$$

$$F = ma$$

$$4000 - (\alpha + 10\beta) = 2000 \times \frac{3}{2}$$

$$\alpha + 10\beta = 1000 \quad \text{--- (1)}$$

$$P = FV$$

$$80 \times 10^3 = F \times 40$$

$$F = 2000$$

$$F = ma$$

$$2000 - (\alpha + 40\beta) = 0$$

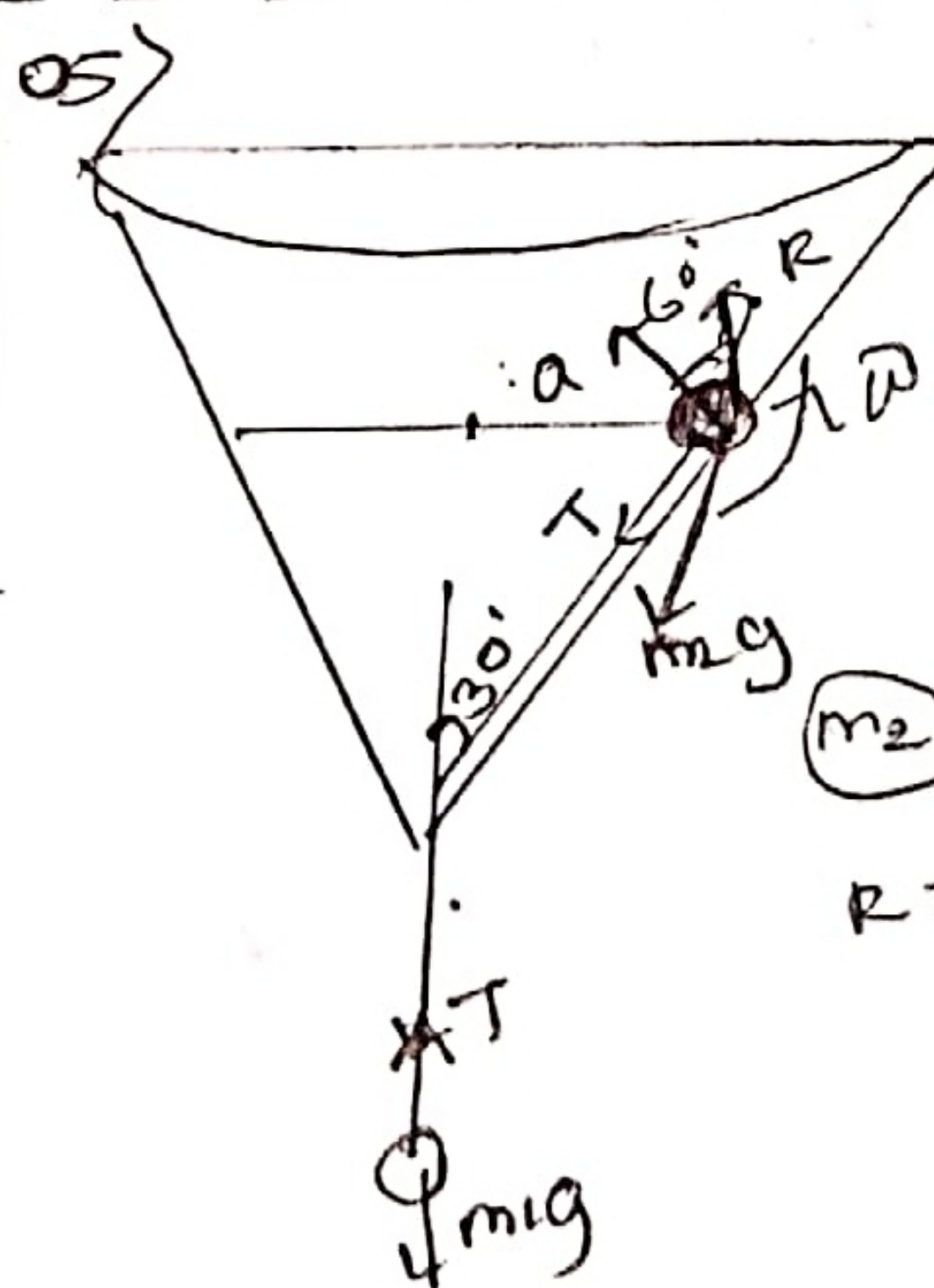
$$\alpha + 40\beta = 2000 \quad \text{--- (2)}$$

$$\text{(2)} - \text{(1)} \quad 30\beta = 1000$$

$$\beta = \frac{100}{3}$$

$$\alpha + 40 \times \frac{100}{3} = 2000$$

$$\alpha = \frac{2000}{3}$$



$$(m_1) \quad T = m_1 g$$

$$(m_2) \quad R \cos 60^\circ - T \cos 30^\circ = m_2 g$$

$$R = (2m_2 + \sqrt{3}m_1)g$$

$$m_2$$

$$R \cos 30^\circ + T \cos 60^\circ = m_2 a \omega^2$$

$$(2m_2 + \sqrt{3}m_1)g \frac{\sqrt{3}}{2} + \frac{Tm_1 g}{2} = m_2 a \omega^2$$

$$(2\sqrt{3}m_2 + 3m_1 + m_1)g = \omega^2$$

$$2m_2 a$$

$$2m_1 + \sqrt{3}m_2 = \omega^2$$

$$2am_2$$

$$\vec{p} = \vec{OA} + \vec{AB}$$

$$= i + 2j + \lambda(i - 3j)$$

$$= (1 + \lambda)i + (2 - 3\lambda)j$$

$$\vec{p} \cdot \vec{AB} = 0$$

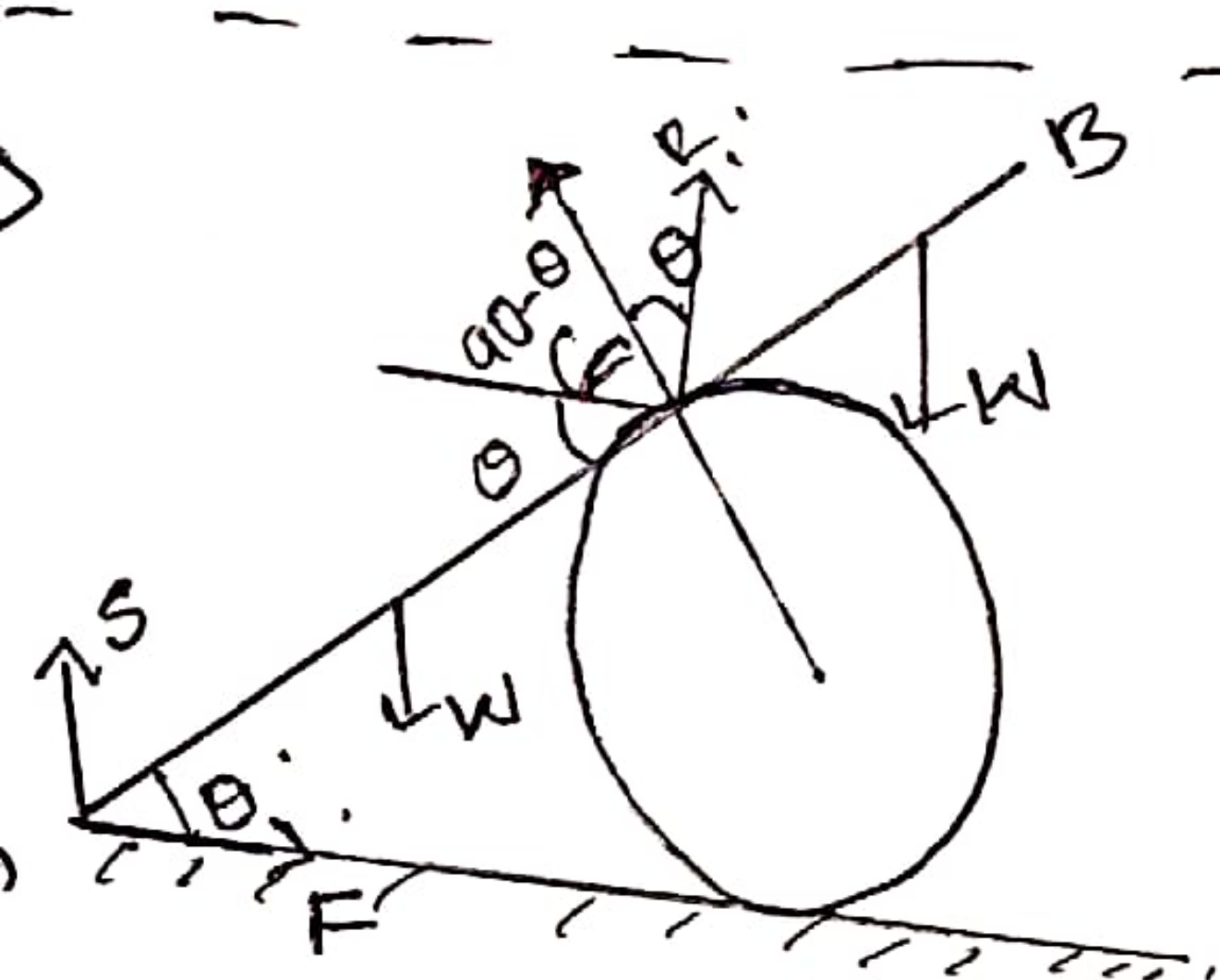
$$(1 + \lambda)i + (2 - 3\lambda)j \cdot [i - 3j] = 0$$

$$1 + \lambda - 3(2 - 3\lambda) = 0$$

$$1 + \lambda - 6 + 9\lambda = 0$$

$$10\lambda = 5$$

$$\lambda = 1/2$$



$$\cos \theta = 3/5$$

$$B) W \cdot 3r \cdot \frac{3}{5} - R \cdot 2r + W \cdot 3r \cdot \frac{3}{5} = 0$$

$$2R = \frac{9W}{10} + \frac{9W}{5}$$

$$2R = \frac{27W}{10}$$

$$\uparrow S - W + \frac{27W}{20} \cdot \frac{3}{5} - W = 0$$

$$S = 2W - \frac{81W}{100}$$

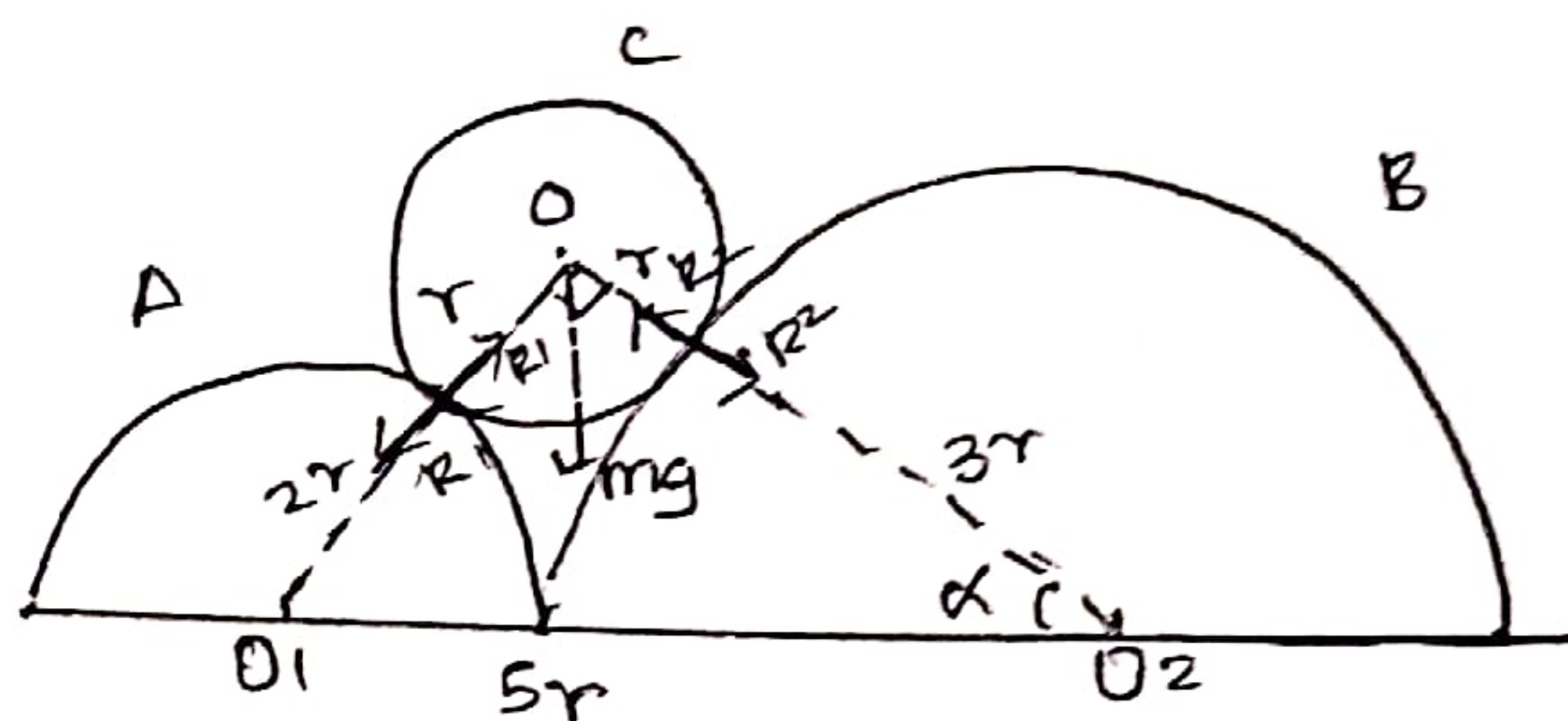
$$S = \frac{119W}{100}$$

$$F = \frac{27W}{20} \times \frac{4}{5}$$

$$= \frac{108W}{100}$$

$$\frac{108W}{119W} \leq \mu \leq \frac{108}{119}$$

08)



Crab

$$\uparrow R_2 \sin \alpha + R_1 \cos \alpha = W$$

$$R_2 \frac{3}{5} + \frac{4}{5} R_1 = W$$

$$3R_2 + 4R_1 = 5W \quad \text{--- (1)}$$

→

$$R_2 \cos \alpha = R_1 \sin \alpha$$

$$R_2 \frac{4}{5} = R_1 \frac{3}{5}$$

$$4R_2 = 3R_1$$

$$R_2 = \frac{3R_1}{4}$$

$$\frac{9R_1}{4} + 4R_1 = 5W$$

$$25R_1 = 20W$$

$$R_1 = \frac{4W}{5}$$

$$R_2 = \frac{3W}{5}$$

$$09) \frac{x}{2} + \frac{x}{4} - \frac{5x}{8} = P(A \cap B)$$

$$= \frac{4x + 2x - 5x}{8}$$

$$P(A \cap B) = P(A) \cdot P(B)$$

$$\frac{x}{8} = \frac{x}{2} \cdot \frac{x}{4}$$

$$x = 1$$

$$(10) 14, 17, 2, 4x - 5y, 19$$

$$\frac{14 + 17 + x + 4x - 5y + 19}{5} = 16$$

$$56 + 5x - 5y = 80$$

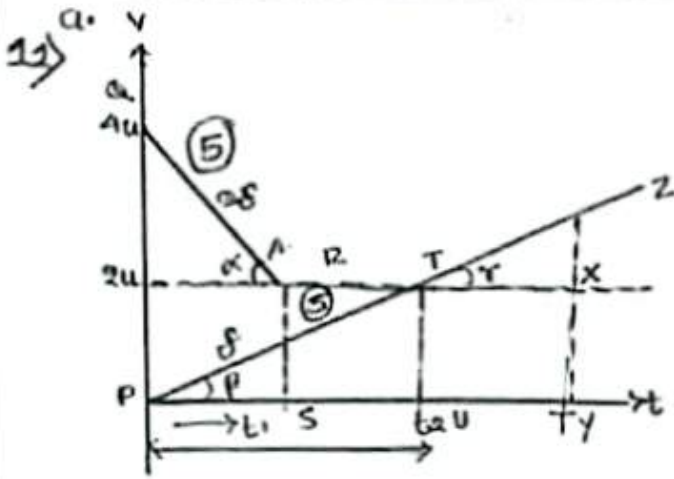
$$5x - 5y = 24 \quad \text{--- (1)}$$

$$\text{Similarly } 4x - 5y = 19 \quad \text{--- (2)}$$

$$x = 11$$

$$11 - y = 6$$

$$y = 5$$



$$\tan \alpha = \frac{2u}{t_1} = \frac{2u}{\frac{4u}{f}} \quad t_1 = \frac{4u}{f} \quad (5)$$

$$t_2 = \tan \beta = \frac{2u}{f} \quad t_2 = \frac{2u}{f} \quad (5)$$

பயணிகள் தொடக்கத்தில் எல்லாம் போகிற,

பயணிகள் தூரம் = $PQRSU$ தூரம் + $RSUT$ தூரம்

$$= \frac{1}{2} \times t_1 \times 2u + (t_2 - t_1) \times 2u$$

$$= \frac{1}{2} \times \frac{4u}{f} \times 2u + \frac{4u}{f} \times 2u$$

$$= \frac{4u^2}{f} + \frac{8u^2}{f} = \frac{12u^2}{f} \quad (5)$$

பயணிகள் தூரம் = ΔPRU தூரம் (5)

$$= \frac{1}{2} \times 2u \times t_2$$

$$= \frac{1}{2} \times 2u \times \frac{2u}{f}$$

$$= \frac{2u^2}{f} \quad (5)$$

$\therefore$ பயணிகள் தூரம் - பயணிகள் தூரம் = d

$$\frac{12u^2}{f} - \frac{2u^2}{f} = d \quad (5)$$

40

பயணிகள் தொடக்கத்தில் தூரம் = $\frac{5u^2}{f} + Tu \times$ தூரம்

$$= \frac{5u^2}{f} + 2u(T - t_2)$$

$$= \frac{5u^2}{f} + 2u(T - \frac{2u}{f})$$

$$= \frac{5u^2}{f} + 2uT - \frac{4u^2}{f}$$

$$= \frac{u^2}{f} + 2uT$$

$$2u = Tf$$

பயணிகள் தொடக்கத்தில் தூரம் = $\frac{1}{2} \times T \times Tf$

$$= \frac{T^2 f}{2} \quad (5)$$

$$\frac{T^2 f}{2} = \frac{5u^2}{f} + 2u(T - \frac{2u}{f}) \quad (5)$$

$$T^2 f^2 = \frac{10u^2}{f} + 4u(Tf - 2u)$$

$$T^2 f^2 = 10u^2 + 4uTf - 8u^2$$

$$T^2 f^2 = 2u^2 + 4uTf$$

$$f^2 T^2 - 4uTf - 2u^2 = 0 \quad (15)$$

$$T = \frac{4u f \pm \sqrt{16u^2 f^2 + 8u^2}}{2f^2} \quad (5)$$

$$= \frac{4u f \pm \sqrt{16u^2 f^2 + 8u^2}}{2f^2}$$

$$= \frac{4u f \pm \sqrt{8u^2(f^2 + 1)}}{2f^2}$$

$$= \frac{4u f \pm \sqrt{8u^2} \sqrt{f^2 + 1}}{2f^2}$$

$$= \frac{4u f \pm 2\sqrt{2}u \sqrt{f^2 + 1}}{2f^2}$$

$$= \frac{2u f \pm \sqrt{2}u \sqrt{f^2 + 1}}{f^2}$$

$T > 0$

$$T = \frac{2u f + \sqrt{2}u \sqrt{f^2 + 1}}{f^2}$$

$$= \frac{2u f + 2\sqrt{2}u \sqrt{f^2 + 1}}{2f^2}$$

$$= \frac{2u f + 2\sqrt{2}u \sqrt{f^2 + 1}}{2f^2}$$

$$= \frac{2u f + 2\sqrt{2}u \sqrt{f^2 + 1}}{2f^2} \quad (5)$$

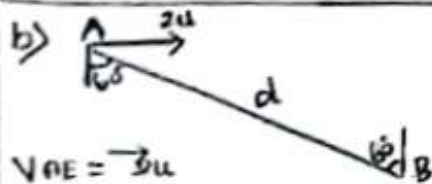
பயணிகள் தொடக்கத்தில் தூரம் = $\frac{T^2 f}{2} \quad (5)$

[தொடக்கத்தில் தூரம்]

$$= \frac{f}{2} \times \frac{(2u f + 2\sqrt{2}u \sqrt{f^2 + 1})^2}{f^4}$$

$$= \frac{f}{2} \times \frac{1}{f^2} (4u^2 f^2 + 8u^2 \sqrt{f^2 + 1} + 8u^2 (f^2 + 1))$$

$$= \frac{f}{2} (5 + 2\sqrt{2}) \quad (15)$$



$$V_{AE} = 3u$$

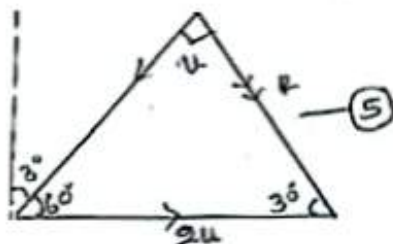
$$V_{BE} = u$$

i) $V_{BE} = u$ (at 30 degrees)

$$V_{AB} = V_{AE} + V_{BE} \quad (5)$$

$$= 3u - \frac{1}{2}u$$

$$= 2u + \frac{1}{2}u$$

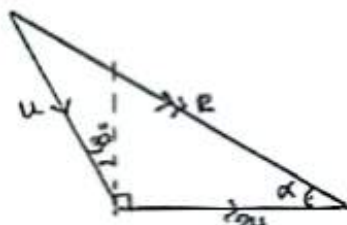


$$R = 3u \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}u}{2} \quad (5)$$

$$T_1 = \frac{d}{V_{AB}} = \frac{d}{\frac{5u}{2}} = \frac{2d}{5u} \quad (5)$$

ii) $V_{BE} = u$

$$V_{AB} = V_{AE} + V_{BE} = 3u + u = 4u \quad (5)$$



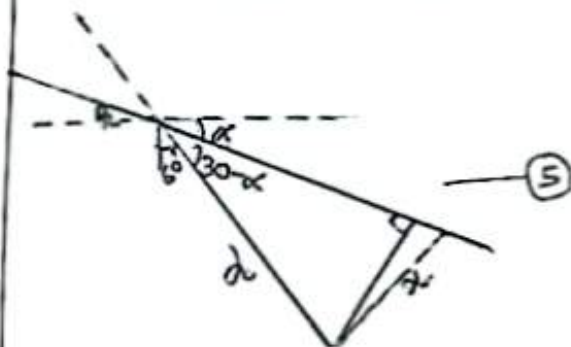
$$\cos(90 + 30) = \frac{4u^2 + u^2 - R^2}{2 \times 4u \times u}$$

$$-\frac{1}{2} = \frac{5u^2 - R^2}{4u^2}$$

$$-2u^2 = 5u^2 - R^2$$

$$R^2 = 7u^2 \quad (5)$$

$$R = \sqrt{7}u$$



විකල්ප 2

$$\frac{\sin \alpha}{u} = \frac{\sin(90 + 30)}{R}$$

$$\sin \alpha = \frac{\sqrt{3}u}{R}$$

$$\sin \alpha = \frac{\sqrt{3}}{2\sqrt{7}} \quad (5)$$

$$x = d \sin(30 - \alpha)$$

$$= d \left[\frac{1}{2} \cos \alpha - \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2\sqrt{7}} \right]$$

$$= d \left[\frac{1}{2} \times \frac{5}{2\sqrt{7}} - \frac{3}{4\sqrt{7}} \right] \quad (5)$$

$$5m'n = \frac{2d}{2\sqrt{7}}$$

$$= \frac{d}{\sqrt{7}} \quad (5)$$

$$T_{min} = \frac{d}{\sqrt{7}u}$$

$$V_{ABC} \text{ වෙලාව } = \sqrt{7}u \times \frac{1}{\sqrt{7}u}$$

$$\tan(30 - \alpha) = \frac{x}{V_{AB}} \quad (5)$$

$$x = \sqrt{7}u \times \frac{1}{2\sqrt{3}}$$

$$\therefore \tan(30 - \alpha) = \frac{1}{2\sqrt{3}}$$

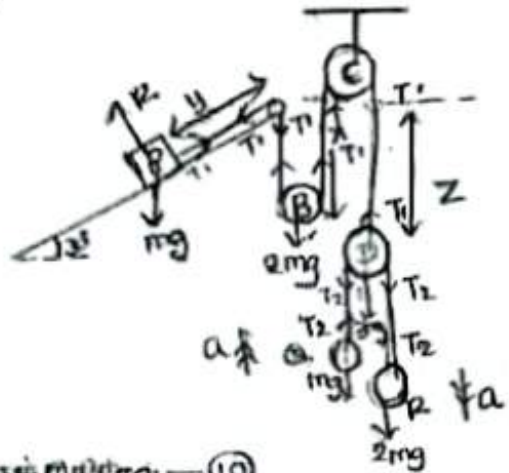
$$x = \frac{\sqrt{7}u}{2\sqrt{3}} \quad (5)$$

$$T_{min} = \frac{d}{2\sqrt{7}} \times \frac{2\sqrt{3}}{\sqrt{7}u}$$

$$= \frac{8\sqrt{3}d}{14u} \quad (5)$$

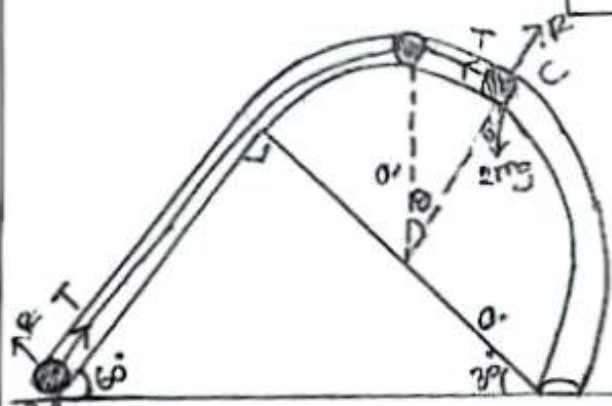
70

12)
9)



$\text{ചിതറൽ തന്മൂലം} \rightarrow (10)$
 $\text{മുഴുവൽ} \rightarrow (20)$
 $\text{A ന്റെ } F = a - \ddot{z} \rightarrow (5) \quad \text{B ന്റെ } F = a - \ddot{z} \rightarrow (5)$
 $y + x + z = 10 \text{ m മാത്രം}$
 $\ddot{y} + \ddot{x} + \ddot{z} = 0 \rightarrow (10)$

$\text{P ന്റെ } F = ma$
 $T_1 - mg \cos 60^\circ = m(-\ddot{y}) \rightarrow (10)$
 $\text{B ന്റെ } \downarrow F = ma$
 $2mg - 2T_1 = 2m\ddot{x} \rightarrow (10)$
 $\text{D ന്റെ } \downarrow 3mg + 2T_2 - T_1 = 3m\ddot{z} \rightarrow (10)$
 $\text{E ന്റെ } 2mg - T_2 = 2m(a + \ddot{z}) \rightarrow (10)$
 $\text{A ന്റെ } T_2 - mg = m(a - \ddot{z}) \rightarrow (10)$



ഊർജ്ജ സംരക്ഷണം
 $0 + 2mg(a + \frac{a}{2}) = \frac{1}{2} \times 2mv^2 + \frac{1}{2}mv^2 + \frac{\sqrt{3}}{2}mg(a\theta) + 2mg(\frac{a}{2}\theta \cos 60^\circ)$
 $3mga = \frac{3}{2}mv^2 + mga\theta + mga\theta \cos 60^\circ + mga\theta \frac{\sqrt{3}}{2}$
 $4ga = 3v^2 + 4ga \cos 60^\circ + \sqrt{3}ga\theta$
 $v = 0$
 $3a\theta^2 = 9[4(1 - \cos 60^\circ) - \sqrt{3}\theta] \rightarrow (10)$

$\text{പ്രകൃത } F = ma$
 $-T + 2mg \sin 60^\circ = 2ma \rightarrow (10)$

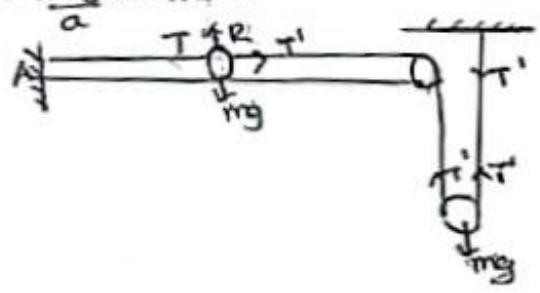
A ന്റെ
 $T - mg \sin 60^\circ = ma$
 $2mg \sin 60^\circ - \frac{mg\sqrt{3}}{2} = 3ma \rightarrow (10)$
 $a = \frac{g}{6} [4 \sin 60^\circ - \sqrt{3}] \rightarrow (*)$

$(4) \text{ തിരുത്തൽ } (2) \text{ തിരുത്തൽ മാത്രമായി}$
 $T = ma + \frac{mg\sqrt{3}}{2}$
 $= \frac{mg}{6} [4 \sin 60^\circ - \sqrt{3}] + \frac{mg\sqrt{3}}{2}$
 $= \frac{mg}{3} (2 \sin 60^\circ + \sqrt{3}) \rightarrow (10)$

$\text{പ്രകൃത } \downarrow F = ma$
 $R - 2mg \cos \theta = -m\frac{v^2}{a}$
 $R = 2mg \cos \theta - \frac{mg}{3} [2(1 - \cos 60^\circ) - \sqrt{3}\theta]$
 $= \frac{10mg \cos \theta}{3} - \frac{4mg}{3} + \frac{\sqrt{3}mg\theta}{3}$
 $R = \frac{mg}{3} [10 \cos \theta - 4 + \sqrt{3}\theta] \rightarrow (10)$

60

$13) \rightarrow$
 $\text{a) } F = ma \rightarrow (10)$
 $T_1 - T = m\ddot{x}$
 $T_1 - \frac{2mg}{a}(x-a) = m\ddot{x}$



$\downarrow F = ma$
 $mg - 2T_1 = m\ddot{y} \rightarrow (5)$
 $mg - 2T_1 = m\frac{\ddot{x}}{2} \rightarrow (5)$

$(1) \times 2 + (2) \rightarrow$
 $\frac{4mg}{a}(x-a) + mg = 6m\ddot{x}$
 $\frac{-4gx + 4ga + ga}{a} = \frac{5\ddot{x}}{2} \rightarrow (15)$

$\frac{-4gx + 5ga}{a} = \frac{5\ddot{x}}{2} \rightarrow (10)$

$\frac{-2}{5} (4gx - 5ga) = \ddot{x}$
 $\frac{-8}{5} g (x - \frac{5a}{4}) = \ddot{x}$
 $\ddot{x} + \frac{8g}{5a} (x - \frac{5a}{4}) = 0 \rightarrow (5)$

$$w = \frac{89}{5a} \quad (5)$$

$$ay + \frac{8}{5a} (ay - \frac{5a}{4}) = 0 \quad (5)$$

$$y = \frac{-4}{5a} (ay - \frac{5a}{4}) \quad (10)$$

$$x - \frac{5a}{4} = A \cos \omega t + B \sin \omega t \quad (15)$$

$$t=0 \quad x=a \quad (5)$$

$$a - \frac{5a}{4} = A + 0$$

$$A = -\frac{a}{4} \quad (5)$$

தகராறு உள்ளதால்

$$x = -A \sin \omega t \cdot \omega + \cos \omega t \cdot \omega \quad (15)$$

$$t=0 \text{ இல் } x=0 \quad (5)$$

$$0 = 0 + B \omega$$

$$B \omega = 0$$

$$\omega = 0 ; \therefore B = 0 \quad (5)$$

$$x - \frac{5a}{4} = -\frac{a}{4} \cos \omega t \quad (5)$$

சமையற்றதில்

$$\cos \omega t = -1 \quad (5)$$

$$x = \frac{5a}{4} + \frac{a}{4} \quad (5)$$

$$= \frac{6a}{4}$$

$$\omega t = \pi$$

$$t = \pi \sqrt{\frac{5a}{8g}} \quad (10)$$

$$t = \frac{\pi}{2} \sqrt{\frac{5a}{2g}} \quad (7.5)$$

14a)

$$(i) \vec{OA} = a, \vec{OB} = b, \vec{OC} = 4a - 3b \quad (5)$$

$$\vec{BA} = \vec{OA} - \vec{OB} = a - b \quad (5)$$

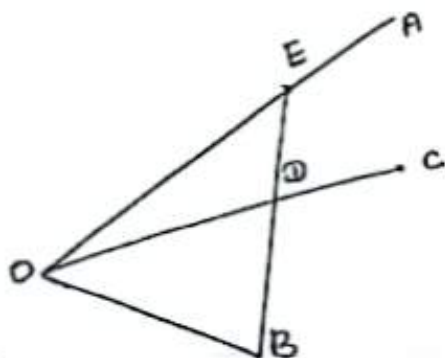
$$\vec{AC} = \vec{OC} - \vec{OA} = 4a - 3b - a = 3a - 3b \quad (5)$$

$$\vec{AC} = 3\vec{BA}$$

$$\therefore A, B, C \text{ ஒரு கோடுகளில் உள்ளன} \quad (5)$$

$$BA : AC = 1 : 3 \quad (5)$$

9.5



$$\vec{OD} = \frac{1}{2} (4a - 3b) \quad (5)$$

$$\vec{OD} = \vec{OB} + \vec{BD}$$

$$= a - \frac{3b}{2} - b$$

$$= a - \frac{5b}{2} \quad (5)$$

(ii) OBE ல் டிரைங்கிள் டிவிட் செய்ய,

$$\vec{OE} = \vec{OB} + \vec{BE}$$

$$\lambda \vec{OA} = b + \mu \vec{BO} \quad (5)$$

$$\lambda a = b + \mu (2a - \frac{5b}{2}) \quad (5)$$

$$\lambda a = b + 2\mu a - \frac{5\mu b}{2}$$

$$\lambda a - 2\mu a - b + \frac{5\mu b}{2} = 0$$

$$a[\lambda - 2\mu] + (\frac{5\mu}{2} - 1)b = 0 \quad (10)$$

$$\lambda - 2\mu = 0 \quad \lambda = 2\mu \quad (5)$$

$$\frac{5\mu}{2} = 1$$

$$\mu = \frac{2}{5} \quad (5)$$

$$\lambda = \frac{4}{5} \quad (5)$$

4.5

$$(ii) \vec{OE} = \frac{4}{5} \vec{OA}$$

$$\vec{OA} - \vec{OE} = \vec{EA}$$

$$\frac{1}{5} \vec{OA} = \vec{EA}$$

$$\vec{OE} : \vec{EA} = 4 : 1 \quad (5)$$

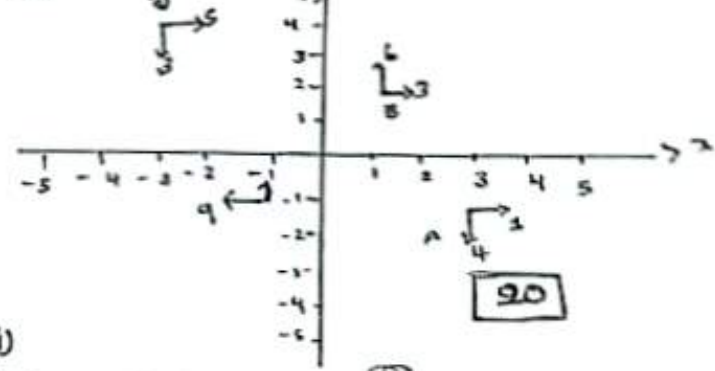
$$\vec{BE} = \frac{2}{5} \vec{BO} \quad (5)$$

$$\vec{BE} : \vec{EO} = 2 : 3 \quad (5)$$

10

(b)

(i)



20

(ii)

$$\uparrow x \Rightarrow 1 + 3 - 9 + 5 = 0 \quad (5)$$

$$\rightarrow y \Rightarrow -4 + 6 + 1 - 3 = 0 \quad (5)$$

$$\begin{aligned} & \vec{OD} = -4x\vec{i} + 1x\vec{j} + 6x\vec{k} + 3x\vec{l} - 1x\vec{i} - 9x\vec{j} + -9x\vec{k} + 3x\vec{l} \\ & = -12 + 1 + 12 - 6 - 1 - 9 - 20 + 9 \\ & = -26 \text{ Nm} \quad (10) \end{aligned}$$

எனவே எதிர்த்து விசையாக செயல்படும் (10)

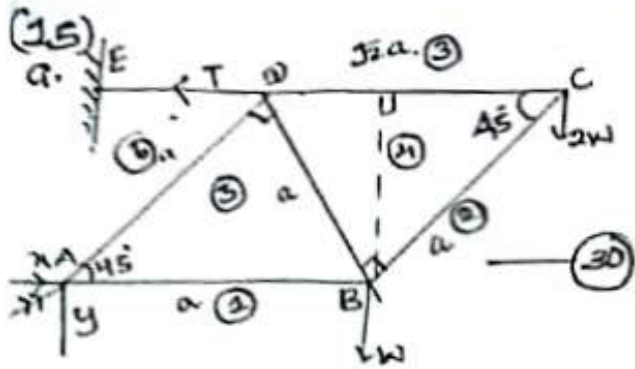
$$(iii) \uparrow x \Rightarrow 1 + 3 - 9 \quad (5)$$

$$\rightarrow y \Rightarrow -4 + 6 + 1 \quad (5)$$

$$z \neq 0 \quad z = -5i + 2j \quad (5)$$

எதிர்த்து விசையாக செயல்படும் (50)

50



$$A) W\sqrt{2}a + 2W(\sqrt{2}a + a \cos 45^\circ) - a \sin 45^\circ T = 0 \quad (10)$$

$$W\sqrt{2}a + 2W(\sqrt{2}a + \frac{a}{\sqrt{2}}) - \frac{a}{\sqrt{2}} T = 0$$

$$\frac{T}{\sqrt{2}a} = 3\sqrt{2}W + \sqrt{2}W \quad (5)$$

$$T = 4\sqrt{2}W \quad (10)$$

$$T = 8W \quad (10)$$

Reaction forces

$$\rightarrow x - 8W = 0 \quad (10)$$

$$x = 8W$$

$$\uparrow -2W - W + y = 0 \quad (10)$$

$$y = 3W$$

$$R = \sqrt{64W^2 + 9W^2}$$

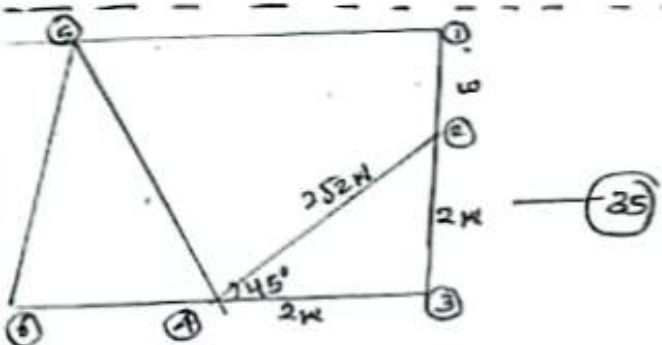
$$= \sqrt{73}W \quad (10)$$

$$\tan \alpha = \frac{3W}{8W}$$

$$= \frac{3}{8}$$

$$\alpha = \tan^{-1}\left(\frac{3}{8}\right) \quad (5)$$

90



Members

BC

CD

BD

AB

AD

Reactions

2W (5)

3\sqrt{2}W (5)

5W (5)

2\sqrt{2}W (5)

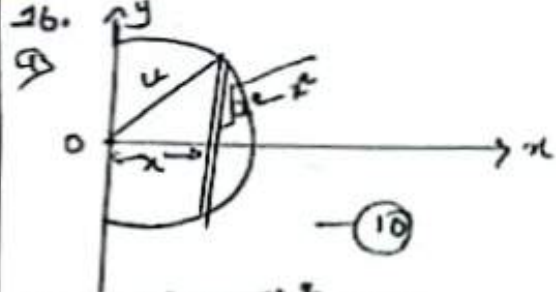
Forces

2\sqrt{2}W (5)

5W (5)

2\sqrt{2}W (5)

60



$$m_i = \pi(\sqrt{a^2 - x^2}) \rho \Delta x$$

$$x_i = x \quad (10)$$

$$\bar{x} = \frac{\int m_i x_i}{\int m_i}$$

$$= \frac{\int_0^a \pi(a^2 - x^2) \rho dx \cdot x}{\int_0^a \pi(a^2 - x^2) \rho dx}$$

$$= \frac{\pi \rho \int_0^a (a^2 - x^2) x dx}{\pi \rho \int_0^a (a^2 - x^2) dx}$$

$$= \frac{\pi \rho \left[\frac{a^2 x^2}{2} - \frac{x^4}{4} \right]_0^a}{\pi \rho \left[a^2 x - \frac{x^3}{3} \right]_0^a}$$

$$= \frac{\pi \rho \left[\frac{a^4}{2} - \frac{a^4}{4} \right]}{\pi \rho \left[a^3 - \frac{a^3}{3} \right]} \quad (20)$$

$$= \frac{\pi \rho \left[\frac{a^4}{4} \right]}{\pi \rho \left[\frac{2a^3}{3} \right]}$$

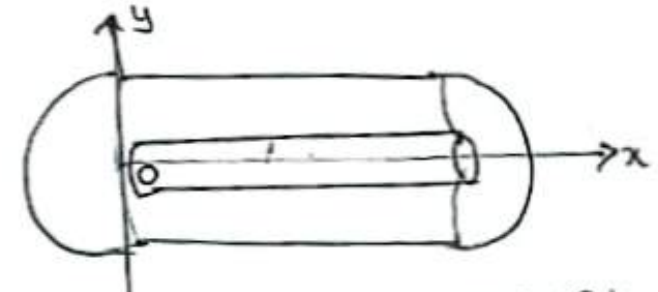
$$= \frac{a^2 x^2 - \frac{x^4}{4}}{a^2 x - \frac{x^3}{3}} \bigg|_0^a$$

$$= \frac{a^2 x - \frac{x^3}{3}}{a^2 x - \frac{x^3}{3}} \bigg|_0^a$$

$$= \frac{a^2 x - \frac{x^3}{3}}{a^2 x - \frac{x^3}{3}} \bigg|_0^a$$

$$= \frac{a^2 x - \frac{x^3}{3}}{a^2 x - \frac{x^3}{3}} \bigg|_0^a$$

40



Volume

(1)

(2)

(3)

(4)

(5)

Area

$\frac{2}{3} \pi a^2 \rho$ (5)

$\pi a^2 \rho$ (5)

$\pi \left(\frac{a}{2}\right)^2 \rho$ (5)

$\frac{16\pi a^3 \rho + 10\pi a^3 \rho - 3\pi a^3 \rho}{12} = \bar{x}$ (5)

$\frac{25}{12} \pi a^3 \rho$ (5)

$\frac{25}{12} \pi a^3 \rho$ (5)

Centroid

$-\frac{30}{8}$ (5)

$\frac{9}{2}$ (5)

$\frac{9}{2}$ (5)

$\frac{16\pi a^3 \rho + 10\pi a^3 \rho - 3\pi a^3 \rho}{12} = \bar{x}$ (5)

$\frac{25}{12} \pi a^3 \rho$ (5)

$\frac{25}{12} \pi a^3 \rho$ (5)

40

2)

$$\frac{25\pi a^3}{12} \rho x = \frac{1}{3} \pi a^3 \rho \left(\frac{3a}{8} \right) + \pi a^3 \rho x - 2\pi \frac{a^3}{4} \rho x \quad \boxed{50V}$$

$$\frac{25}{6} \pi x = -a + 4 - a/4$$

$$x = \frac{-3a}{50}$$

$$Q = \frac{4}{3} \pi a^3 \rho x - \frac{3a}{8} + 2a^3 \rho x \frac{a}{2} - \frac{\pi a^3 \rho a}{4} \quad \text{--- (16)}$$

$$= \frac{\pi a^3 \rho}{4} h \frac{h}{2}$$

$$= -a^2 g + a^2 g = \frac{a^2 g}{4} + \frac{h^2 g}{4}$$

$$\frac{g}{8} = \left(\frac{h}{a} \right)^2$$

$$\boxed{30}$$

(27) எந்தெந்தவற்றை உபயோகிப்போம் = 6x6x6

(d) தரப்பட்டவை n

$$P(A=1) = \frac{3}{6} = \frac{1}{2}$$

$$P(A=2) = \frac{2}{6} = \frac{1}{3} \quad \text{--- (15)}$$

$$P(A=3) = \frac{1}{6}$$

தரப்பட்டவை B

$$P(B=1) = \frac{1}{6}$$

$$P(B=2) = \frac{2}{6} = \frac{1}{3} \quad \text{--- (15)}$$

$$P(B=3) = \frac{1}{2}$$

$$\boxed{35}$$

i) எந்தெந்தவற்றை ஒத்தவைகள்/பெயர்வைகள்
என்பதை கவனத்தில் கொள்ளுங்கள்.

உதாரணமாக: 1x1x1, 2x2x2, 3x3x3

$$P(A=1) + P(A=3) = \frac{3}{6} + \frac{1}{6} = \frac{4}{6} \quad \text{--- (5)}$$

$$P(B=1) + P(B=3) = \frac{1}{6} + \frac{1}{2} = \frac{4}{6} \quad \text{--- (6)}$$

$$P(C) = \frac{1}{4} = \frac{1}{4} \quad \text{--- (5)}$$

உதாரணமாக: 1x1x1, 2x2x2, 3x3x3

$$P(A=2) = \frac{2}{6} = \frac{1}{3} \quad \text{--- (5)}$$

$$P(C=2) = \frac{1}{6}$$

$$\frac{20}{36} = \frac{5}{9} \quad \text{--- (5)}$$

ii) எந்தெந்தவற்றை (A=1, B=2), (A=2, B=1)

$$P((A=1), (B=2)) = \frac{1}{6} \times \frac{2}{6} = \frac{1}{18}$$

$$P(A=2, B=1) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36} \quad \text{--- (5)}$$

$$iii) P(A) = \frac{1}{3}$$

$$P(A=2, B=1) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36} \quad \text{--- (5)}$$

$$\boxed{35}$$

(28)
a)



வருமானம்

முழுது

செலவுகள்



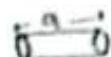
$$\frac{2}{3} \pi a^2 g$$

$$= \frac{2a}{3}$$



$$2a^2 g$$

$$\frac{1}{2}$$



$$- \frac{\pi a^2 g}{4}$$

$$\frac{1}{2}$$



$$\frac{\pi a^2 h^2}{4}$$

$$\frac{h}{2}$$



$$\frac{25\pi a^3 g + \pi a^2 h^2}{12}$$

$$0$$